pyAmpliCol 0.1.0

Methodology and Performance Report

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Release 0.1.0

Measurement profile: `x86_EPYC_manual`

Abstract

This report presents the physics coverage, numerical validation, and performance of `pyAmpliCol` 0.1.0. We compare compiled, eager, and recurrence evaluation for the built-in Standard Model and for the same model imported from UFO or serialized JSON, together with scalar-contact and scalar-gravity benchmarks. Every reported timing is associated with a documented software revision, hardware profile, numerical configuration, and validation reference appropriate to the corresponding data set.

Contents

1 Scope

`pyAmpliCol` generates and evaluates tree-level scattering amplitudes. Compiled mode specializes a shared-current directed acyclic graph (DAG) to a process. Eager mode evaluates the same graph with reusable model kernels. Recurrence mode constructs a compact current schedule directly. The three modes implement the same physics normalization, colour treatment, helicity sums, and resolved observables. The original `AmpliCol` program and its use in event generation are described in Ref. [5].

The report has three purposes:

1. define the observables and conditions required for a reproducible benchmark;
2. preserve a stable matrix of representative process families and colour approximations; and
3. present performance only after numerical agreement has been verified.

In each architecture-specific report, an N/A entry denotes a process/multiplicity combination outside the declared scope or an observable that does not apply. Every applicable cell across the complete declared multiplicity range must instead contain a validated measurement or an documented resource-limit status. A measurement that does not pass numerical validation is labelled explicitly and never replaced by a timing value. No value is copied, interpolated, or estimated from a neighbouring cell.

2 Current Recursion and Model Lowering

2.1 Shared-current DAG

For an external subset I, particle state t, and local spin or flow label χ, an off-shell current has the schematic recursion

$$J^{\alpha}_{t,\chi}(I) = P^{\alpha}{}_{\beta}(t, p_I) \sum_{I=A\dot{\cup}B} V^{\beta}{}_{\gamma\delta}(t_A, t_B \to t) J^{\gamma}_{t_A,\chi_A}(A) J^{\delta}_{t_B,\chi_B}(B), \qquad p_I = \sum_{i\in I} p_i. \tag{1}$$

Current identity includes the particle, external labels, helicity ancestry, chirality or spin state, flavour and charge flow, colour state, momentum mask, and any auxiliary-current kind. Contributions with the same complete identity share one node. A topological sweep therefore evaluates each retained current once per phase-space point and reuses it in all dependent amplitudes. This is a Berends–Giele-type construction [1].

The Universal Feynman Output format (UFO; originally Universal FeynRules Output) describes particles, parameters, interactions, Lorentz tensors, colour tensors, and propagators in a model-independent form [3, 4]. `pyAmpliCol` normalizes a trusted UFO Python module, or its own serialized JSON representation of that model, to the same canonical representation used by process generation. The UFO module is executable Python; the corresponding serialized JSON form is data-only.

Optimizations of imported models are based on exact colour, Lorentz, propagator, and coupling identities rather than particle names or numerical codes. For Standard-Model validation processes, the built-in and imported routes must produce equivalent current, interaction, amplitude, and colour content. Algebraic reuse is allowed only when the relevant identity has been established exactly.

2.2 Contact terms and spin two

A supported vertex with valence above three is lowered to a deterministic tree of non-propagating auxiliary currents. The construction inserts the original coupling once and introduces no pole. Direct contraction of the original contact tensor is the validation reference for the scalar-contact ladder.

The scalar-gravity ladder exercises rank-two external states and momentum-dependent interactions. For a massless graviton, the helicity-two polarization tensor is represented as

$$\epsilon^{\mu\nu}_{\pm 2} = \epsilon^{\mu}_{\pm}\epsilon^{\nu}_{\pm}. \tag{2}$$

Validation checks symmetry, transversality, tracelessness, normalization, and agreement between ordinary and higher-precision evaluation.

3 Colour and Runtime Semantics

The process matrices use three colour contracts:

LC	Leading-colour ordered amplitudes with physical colour-flow selectors [2].
NLC	Every colour-metric entry whose leading power is at most two below the leading power of the complete basis.
Full	The complete supported sparse colour metric for the generated amplitude basis.

For NLC, "retained" applies to a colour-matrix *entry*, not to a truncated numerical coefficient. Write the exact overlap of two colour-flow tensors as

$$C_{ab}(N_c) = \sum_p c_p^{(ab)} N_c^p, \qquad p_{ab} = \max\{p : c_p^{(ab)} \neq 0\}, \tag{3}$$

and let P be the leading power of the complete colour basis. The sparse NLC metric is

$$C_{ab}^{\mathrm{NLC}} = \begin{cases} C_{ab}(3), & p_{ab} \geq P-2, \\ 0, & p_{ab} < P-2. \end{cases} \tag{4}$$

Thus every qualifying entry keeps all of its finite-N_c terms. For three open fundamental lines and one adjoint, $P = 4$. Representative retained off-diagonal overlaps are

exact overlap	highest power	NLC at $N_c = 3$	full at $N_c = 3$
$-N_c^3 + N_c$	3	−24	−24
$N_c^2 - 1$	2	8	8

The powers and coefficients are retained exactly. When only the upper-triangular part of the colour metric is stored, off-diagonal interference terms receive the usual factor of two; this factor is not part of the colour coefficient in Eq. (4).

The summed observable returns one matrix element per point. Leading-colour resolved values have axes `(point, helicity, flow)`; NLC and full-colour values have one contracted colour component per helicity. Summing all non-point axes must reproduce the directly evaluated total. Selectors refer to physical helicities and flows.

All `pyAmpliCol` binary64 interfaces use the same native numerical core. Cross-interface agreement therefore checks data conventions, while a separately generated value is required for an independent physics comparison.

The JIT backend stores a portable SymJIT application for binary64 evaluation; the receiving host creates code appropriate for its processor. ASM and C++ payloads are target-specific. Evaluation above binary64 precision uses the retained symbolic evaluator state.

4 Benchmark Methodology

4.1 Configuration contract

Each measurement begins from an immutable resolved configuration. Unless a table states otherwise, measurements use a batch size of 128, two warmup runs, at least five timed samples, a five-second sampling target, and one CPU core. Compiled process matrices use JIT O3; eager and recurrence matrices use portable JIT O2 prepared models. Explicitly labelled rows in the dedicated Z ladder use JIT O1, JIT O3, ASM O3, or C++ O3. Backend-specific optimization settings are retained with the measurement record.

Generation time begins with the resolved model input required by an execution mode and ends with a validated, loadable process representation. Compiled mode includes process-specific evaluator construction and backend compilation. Eager and recurrence modes include construction and serialization of their respective numerical schedules. Preparation of a reusable built-in or UFO/JSON model bundle occurs before this timer and is therefore excluded from the process-generation values in the tables.

Runtime measurements use the summed matrix element. The reported native wall time starts after caller-language array conversion and includes source construction, numerical evaluation, selector handling, and final reduction. Caller-language packing is excluded. When a backend exposes a separately measured execution attribution, the tables show it alongside wall time; otherwise the field is marked NOT EXPOSED and is never inferred or replaced by zero. Comparisons use the same process, colour accuracy, parameters, phase-space point, precision, selectors, and normalization.

4.2 Benchmark platform

This document is an architecture-specific edition. The recorded hardware, operating system, and processor summary is pending measurement-host authentication. The measurement profile is `x86_EPYC_manual`; its recorded software environment is pending an authenticated post-checkpoint build. Measurements from different processor architectures are reported separately and are never combined in a summary ratio. Unless a table states otherwise, generation and evaluation use one CPU core and a five-second sampling target.

4.3 Numerical validation

For each Standard-Model process, all compared execution modes and the original `AmpliCol` reference are evaluated at the same deterministic phase-space point with identical numerical parameters. Cross-mode comparisons use relative tolerance 10^{-12} and absolute tolerance 10^{-15}; comparisons with the independent `AmpliCol` result use relative tolerance 10^{-8}. Scalar-model ladders instead compare binary64 evaluation with higher precision and, where applicable, direct contact-tensor contraction. Every result must reproduce its resolved sum and preserve expected structural zeros.

Timing and numerical validation are separate requirements: a timing is reported only when the corresponding matrix element passes all validation checks, and numerical agreement alone does not substitute for a measured generation or runtime value.

5 Standard-Model Process Matrices

The process matrices use recurrence as the primary `pyAmpliCol` execution mode. Built-in-SM and UFO-SM recurrence JIT O2 are compared with independent original-`AmpliCol` results. Separate built-in-SM matrices compare compiled JIT O3 and eager JIT O2 with recurrence JIT O2 for the same process, colour treatment, and observable.

The canonical matrix spans electroweak vector production, charged-current and leptonic channels, heavy-quark production, pure gluons, multi-boson final states, Higgs-associated final states, and three- and four-quark-line cases. Here n is the number of final-state particles. LC coverage retains every declared family through $n = 9$. NLC and full-colour grids retain $n = 1, \ldots, 5$. A process/multiplicity combination outside its family definition is marked NOT APPLICABLE; this is table structure, not an unfilled measurement. A complete edition accounts for every applicable cell across the full declared range. Resource-limited cells carry a documented status; every successful cell is numerical and validated. An in-scope result that does not pass numerical validation is identified by a status and does not carry a timing ratio.

The UFO-SM recurrence matrices use the same process definitions, colour contracts, numerical parameters, selectors, and validation points as the built-in-SM recurrence matrices. Their source is the packaged Standard Model loaded through the serialized-JSON interface. This comparison verifies that the imported and built-in model routes implement the same requested physics. Compiled and eager process matrices are built-in-only, isolating differences between execution modes from differences in model loading.

Numerical validation summary

Only measurements whose numerical checks pass are counted below. The scope comprises every applicable report cell across the complete declared multiplicity range.

final-state multiplicity	declared	static N/A	verified	measurable coverage
1	64	0	64	verified
2	186	0	186	verified
3	206	0	206	verified
4	306	0	306	verified
5	305	0	305	verified
6	205	4	196	incomplete (0 censored, 5 open)
7	175	10	156	incomplete (0 censored, 9 open)
8	175	10	40	incomplete (0 censored, 125 open)
9	174	10	26	incomplete (0 censored, 138 open)
total	1796	34	1485	incomplete (0 censored, 277 open)

Full-catalog display accounting through $n = 9$: 1796 declared cells, comprising 1762 measurable cells and 34 catalog-authenticated static N/A rows; 432 matrix process/multiplicity positions marked either NOT APPLICABLE or NOT RUN; and 36 reference execution fields marked NOT EXPOSED. The latter two categories are intentional table structure, not unfilled measurements.

validation comparison	passed	max. relative difference
independent original-`AmpliCol` reference records	272	reference
`pyAmpliCol` versus independent reference	508	1.123×10^{-11}
compiled/eager versus recurrence	500	3.328×10^{-13}
optimized result versus resolved sum	1213	3.254×10^{-13}
scalar result versus higher precision	25	2.384×10^{-14}

Profile: `x86_EPYC_manual`. Measured source: source identity is incomplete or mixed. Independent-reference comparisons use a relative tolerance of 10^{-8}; cross-mode, resolved-sum, and higher-precision comparisons use 10^{-12}.

Workloads and denominators

For a reported timing quantity T, each multiplier is

$$r_{\text{candidate/baseline}} = \frac{T_{\text{candidate}}}{T_{\text{baseline}}}. \tag{5}$$

Green, orange, and red denote respectively $r < 1$, $1 \leq r < 2$, and $r \geq 2$. Green therefore means that the candidate is faster than the baseline named in the table heading. The recurrence matrices use original `AmpliCol` as the baseline. The compiled and eager matrices use built-in-SM recurrence JIT O2 as the baseline.

Every LC comparison contains two complete-coverage layouts:

- topology replay, measured with one physical colour flow selected and all retained helicities summed; and
- the all-flow union, measured with one physical helicity selected and all retained colour flows summed.

Neither selection is fixed during generation. Each LC cell reports the two baseline generation quantities and wall times, followed by candidate generation entries and runtime ratios. The muted supplementary ratio first compares separately measured execution attributions when both sides expose compatible boundaries; otherwise it compares authenticated evaluator totals. Only against original `AmpliCol`, which has no evaluator-total clock, may a candidate evaluator total use the legacy direct execution clock (or wall when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. If no compatible supplementary clock exists, only the wall ratio is shown. Compiled and eager generation are compared with recurrence using the same layout.

The two original-`AmpliCol` reference workloads have different setup boundaries. For the selected-flow workload, the reported generation quantity is creation of the process library. For the all-flow workload, it is preparation of direct all-flow evaluation. Runtime ratios compare the same physical observable. The all-flow setup quantities are not like-for-like, so the table shows the candidate process-generation time as an absolute value and forms no ratio.

NLC and full colour use one helicity-summed, colour-contracted calculation per mode. Their cells report the absolute baseline generation and wall time, followed by the candidate generation and runtime ratios. No per-flow scalar is defined for these contracted colour accuracies.

The wall quantity is the repeated native evaluation time after caller-language input conversion and is the primary runtime observable. Independent accumulated evaluator-total and narrower recurrence-core clocks remain in the detailed measurement evidence; neither is copied from wall time or from the other. The compact runtime notation `([xS]xW)` shows a compatible supplementary clock multiplier `xS` in muted brackets and the primary wall multiplier `xW` in its performance colour. The hierarchy above determines whether `xS` is execution attribution or a diagnostic evaluator-total fallback. If no supplementary ratio is eligible, the cell shows only `(xW)`.

Preparation of reusable model-level kernels is excluded from process-generation time. Compiled generation still includes process-specific evaluator construction and backend compilation.

A timing is shown only after recurrence agrees with original `AmpliCol` at the independent-reference tolerance and compiled and eager results agree with recurrence at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are also compared directly at that stricter tolerance, rather than relying on their separate agreement with `AmpliCol`. Each result must reproduce its resolved sum. For LC, the common flow–helicity component is compared directly between topology replay and all-flow union, and every `pyAmpliCol` all-flow result is also compared with the matching independently generated `AmpliCol` component. A failed validation has no timing ratio.

Cell and summary layout

<table>
<tr><th>contract</th><th>baseline entry</th><th>candidate entries</th></tr>
<tr><td>LC, upper</td><td>topology replay <code>|</code> all-flow generation quantities in seconds</td><td>layout-matched ratios, except that the all-flow candidate against original <code>AmpliCol</code> is shown absolutely without a ratio</td></tr>
<tr><td>LC, lower</td><td>selected-flow <code>|</code> all-flow wall times in μs/pt</td><td>compact <code>([xS]xW)</code> comparisons, or <code>(xW)</code> when only wall is comparable</td></tr>
<tr><td>NLC and full-colour, upper</td><td>contracted generation time in seconds</td><td>candidate generation ratio</td></tr>
<tr><td>NLC and full-colour, lower</td><td>contracted wall time in μs/pt</td><td>compact <code>([xS]xW)</code> comparison, or <code>(xW)</code> when only wall is comparable</td></tr>
</table>

Summary rows contain multipliers only, ordered as minimum, maximum, median, arithmetic mean, and weighted mean. The arithmetic mean averages the individual candidate/baseline multipliers; the weighted mean is the ratio of sums $\sum_i T_{\mathrm{candidate},i} / \sum_i T_{\mathrm{baseline},i}$. Generation summaries use topology replay for LC and the contracted workload for NLC and full colour; the non-comparable LC all-flow generation quantity is omitted. Runtime summaries use wall time only, with separate aligned LC topology-replay and all-flow lines. Structurally inapplicable, resource-limited, unsupported, and validation-failed cells do not contribute.

The first three matrices provide a built-in-SM overview against original `AmpliCol`. For each process, multiplicity, colour treatment, and LC workload, they select the validated pyAmpliCol mode with the smallest wall time. Selection is independent in every cell: the overview records the measured winner rather than prescribing one mode for all processes. Its upper row reports generation and its lower row reports runtime. LC pairs use non-union-flow `|` union-flow ordering. Since original-`AmpliCol` all-flow setup and `pyAmpliCol` process generation have different boundaries, the latter is shown absolutely rather than as a misleading ratio.

5.1 Built-in SM best measured mode versus AmpliCol: LC

Built-in SM best measured mode versus AmpliCol LC (block 1 of 3)

ID	base process	metric	n=1				n=2				n=3			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.14	0.000973	(r) x1.16	8.12 (r)	7.15	0.00107	(r) x1.25	8.21 (r)	7.32	0.000760	(c) x0.932	4.85 (c)
		run [μs/pt]	0.0561	0.122	([x3.05] x4.36)	([x0.185] x1.17)	0.362	0.280	([x1.47] x1.83)	([x0.377] x1.07)	1.23	0.770	([x1.00] x1.00)	([x0.690] x0.690)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.72	0.000448	(r) x1.04	8.05 (r)	7.89	0.000850	(r) x1.07	9.16 (r)	7.61	0.000766	(c) x0.874	4.41 (c)
		run [μs/pt]	0.0323	0.124	([x3.82] x5.81)	([x0.171] x1.10)	0.219	0.284	([x1.66] x2.23)	([x0.366] x1.05)	0.752	0.748	([x1.53] x1.53)	([x0.703] x0.703)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE				9.36	0.00106	(r) x0.872	8.27 (r)	9.52	0.00149	(r) x0.927	5.13 (c)
		run [μs/pt]					0.185	0.268	([x1.37] x1.88)	([x0.384] x0.904)	0.453	0.430	([x1.16] x1.51)	([x0.885] x0.885)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE				9.18	0.000491	(r) x0.930	8.47 (r)	9.31	0.000688	(r) x0.892	8.75 (r)
		run [μs/pt]					0.0665	0.212	([x1.50] x2.77)	([x0.343] x1.01)	0.139	0.363	([x1.39] x2.40)	([x0.417] x0.999)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE				9.90	0.000496	(c) x0.484	8.40 (r)	10.1	0.000726	(c) x0.560	4.35 (c)
		run [μs/pt]					0.469	0.295	([x1.83] x1.83)	([x0.356] x1.04)	2.07	0.605	([x0.981] x0.982)	([x0.814] x0.815)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.90	0.000518	(c) x0.549	3.91 (c)	17.0	0.000889	(c) x0.610	5.58 (c)
		run [μs/pt]					0.319	0.416	([x2.29] x2.29)	([x0.895] x0.895)	1.95	1.62	([x0.963] x0.964)	([x0.669] x0.669)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				9.22	0.00106	(r) x0.892	8.96 (r)	13.3	0.000754	(c) x0.570	5.68 (c)
		run [μs/pt]					0.146	0.283	([x1.96] x2.76)	([x0.289] x0.905)	0.625	0.918	([x1.60] x1.60)	([x0.654] x0.655)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE				11.1	0.000548	(c) x0.485	4.22 (c)	13.1	0.00217	(c) x0.865	6.74 (c)
		run [μs/pt]					0.459	0.596	([x1.40] x1.40)	([x0.731] x0.731)	2.39	3.67	([x0.853] x0.854)	([x0.514] x0.514)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				19.3	0.00147	(c) x0.376	4.64 (c)
		run [μs/pt]									4.57	0.849	([x0.942] x0.942)	([x0.705] x0.705)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												

	n=1 [r:2|c:0|e:0]					n=2 [r:5|c:3|e:0]					n=3 [r:2|c:7|e:0]				
summary: generation	x1.04	x1.16	x1.10	x1.10	$\boxed{\text{x1.10}}$	x0.484	x1.25	x0.882	x0.816	$\boxed{\text{x0.789}}$	x0.376	x0.932	x0.865	x0.734	$\boxed{\text{x0.683}}$
summary: run single-flow, hel. sum	x4.36	x5.81	x5.09	x5.09	$\boxed{\text{x4.89}}$	x1.40	x2.77	x2.06	x2.12	$\boxed{\text{x1.94}}$	x0.854	x2.40	x1.00	x1.31	$\boxed{\text{x1.03}}$
summary: run all-flows, single hel.	x1.10	x1.17	x1.13	x1.13	$\boxed{\text{x1.13}}$	x0.731	x1.07	x0.957	x0.951	$\boxed{\text{x0.921}}$	x0.514	x0.999	x0.703	x0.737	$\boxed{\text{x0.648}}$

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol LC (block 2 of 3)

ID	base process	metric	n=4				n=5				n=6			
1	$dd \to Z + (n-1)g$	gen. [s]	7.84	0.00489	(c) x1.58	9.30 (c)	8.16	0.0149	(c) x4.60	30.7 (c)	8.78	0.140	(c) x26.2	17.9 (r)
		run [μs/pt]	4.46	3.05	([x0.724] x0.724)	([x0.447] x0.447)	13.2	15.3	([x0.752] x0.752)	([x0.457] x0.457)	34.1	96.3	([x1.11] x1.11)	([x0.661] x0.685)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.83	0.00238	(c) x1.47	8.07 (c)	8.15	0.0149	(c) x3.46	19.9 (c)	8.40	0.149	(r) x1.60	19.2 (r)
		run [μs/pt]	2.52	4.21	([x0.866] x0.866)	([x0.366] x0.366)	9.00	15.4	([x0.783] x0.783)	([x0.448] x0.448)	26.6	98.6	([x0.651] x0.689)	([x0.687] x0.706)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.42	0.00201	(c) x0.966	6.71 (c)	9.89	0.0115	(c) x1.80	13.5 (c)	11.2	0.0978	(c) x4.80	44.6 (c)
		run [μs/pt]	1.24	0.964	([x0.945] x0.946)	([x0.656] x0.657)	3.85	3.70	([x0.767] x0.768)	([x0.423] x0.423)	11.1	16.7	([x0.809] x0.809)	([x0.458] x0.458)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.29	0.00197	(r) x0.955	4.48 (c)	9.41	0.0114	(c) x0.994	7.54 (c)	10.1	0.0971	(c) x2.35	19.5 (c)
		run [μs/pt]	0.413	0.831	([x1.19] x1.76)	([x0.728] x0.728)	1.66	3.15	([x1.02] x1.02)	([x0.471] x0.471)	6.05	15.2	([x0.860] x0.860)	([x0.475] x0.475)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	0.00433	(c) x0.933	6.27 (c)	11.5	0.0123	(c) x1.90	13.5 (c)	13.4	0.102	(r) x1.17	60.2 (c)
		run [μs/pt]	6.36	1.68	([x0.694] x0.694)	([x0.535] x0.535)	16.7	6.30	([x0.785] x0.785)	([x0.394] x0.394)	42.9	30.6	([x0.804] x0.846)	([x0.388] x0.388)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	30.8	0.00328	(c) x1.18	18.2 (c)	64.5	0.0348	(r) x1.15	13.7 (r)	186	0.296	(r) x1.30	59.9 (r)
		run [μs/pt]	6.89	8.41	([x0.896] x0.896)	([x0.860] x0.860)	21.2	77.5	([x1.09] x0.888)	([x0.478] x0.861)	58.0	479	([x0.932] x1.07)	([x0.969] x1.02)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	17.6	0.00499	(c) x1.20	10.9 (c)	23.6	0.0146	(c) x2.64	37.4 (c)	33.2	0.137	(r) x1.31	236 (c)
		run [μs/pt]	2.39	4.16	([x0.943] x0.943)	([x0.449] x0.449)	8.08	23.6	([x0.899] x0.900)	([x0.437] x0.437)	23.6	157	([x0.859] x0.934)	([x0.624] x0.624)
8	$gg \to gg + (n-2)g$	gen. [s]	15.8	0.00514	(r) x1.15	24.2 (c)	18.3	0.0838	(r) x2.87	37.3 (r)	27.8	3.09	(r) x4.83	2.87e+03 (c)
		run [μs/pt]	9.36	26.2	([x0.712] x0.760)	([x0.612] x0.612)	28.7	221	([x0.650] x0.682)	([x0.757] x0.775)	79.0	2.74e+03	([x0.638] x0.666)	([x0.826] x0.826)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.3	0.00206	(c) x0.504	5.89 (c)	26.8	0.0112	(r) x0.518	8.85 (c)	38.4	0.0927	(r) x0.557	31.3 (c)
		run [μs/pt]	12.9	1.56	([x0.918] x0.919)	([x0.608] x0.608)	33.3	3.94	([x0.848] x0.894)	([x0.451] x0.451)	84.1	14.5	([x0.806] x0.841)	([x0.344] x0.344)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	57.0	0.00195	(c) x0.140	6.19 (c)	61.7	0.0111	(c) x0.164	7.39 (c)	65.9	0.0922	(c) x0.402	11.0 (c)
		run [μs/pt]	2.06	1.29	([x1.01] x1.01)	([x0.608] x0.609)	4.66	1.92	([x0.863] x0.864)	([x0.568] x0.568)	10.7	3.90	([x0.797] x0.797)	([x0.475] x0.475)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	31.1	0.00201	(c) x0.284	6.08 (c)	64.4	0.0104	(c) x0.373	10.1 (c)	117	0.107	(r) x0.259	33.5 (c)
		run [μs/pt]	2.99	1.63	([x0.991] x0.992)	([x0.602] x0.603)	9.01	4.83	([x0.999] x0.999)	([x0.526] x0.526)	26.0	36.5	([x1.12] x1.19)	([x0.278] x0.278)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	63.3	0.00432	(c) x0.205	11.5 (c)	67.5	0.0110	(c) x0.252	14.0 (c)	65.6	0.0913	(c) x0.622	23.4 (c)
		run [μs/pt]	3.81	1.85	([x0.580] x0.580)	([x0.481] x0.481)	7.86	2.74	([x0.542] x0.542)	([x0.460] x0.460)	17.8	4.96	([x0.488] x0.488)	([x0.422] x0.422)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	12.4	0.00149	(c) x1.09	10.5 (c)	21.5	0.00985	(c) x1.89	27.1 (c)	31.6	0.0946	(c) x8.23	115 (c)
		run [μs/pt]	0.457	2.07	([x1.86] x1.86)	([x0.507] x0.507)	1.14	9.33	([x1.35] x1.35)	([x0.412] x0.412)	3.05	50.9	([x0.969] x0.970)	([x0.421] x0.421)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				STATIC N/A	STATIC N/A	132 (c)	95.3 (c)
		run [μs/pt]									STATIC N/A	STATIC N/A	5.01	10.8
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	11.9	0.00195	(c) x2.11	15.6 (c)	18.7	0.0110	(c) x3.94	42.5 (c)	26.4	0.101	(c) x11.3	212 (c)
		run [μs/pt]	0.678	3.22	([x1.43] x1.43)	([x0.483] x0.483)	1.58	15.8	([x1.08] x1.08)	([x0.408] x0.408)	3.90	87.2	([x0.926] x0.927)	([x0.459] x0.459)

	n=4 [r:2|c:12|e:0]					n=5 [r:3|c:11|e:0]					n=6 [r:7|c:7|e:0]				
summary: generation	x0.140	x2.11	x1.03	x0.983	**x0.675**	x0.164	x4.60	x1.84	x1.90	**x1.17**	x0.259	x26.2	x1.45	x4.63	**x2.22**
summary: run single-flow, hel. sum	x0.580	x1.86	x0.931	x1.03	**x0.853**	x0.542	x1.35	x0.876	x0.879	**x0.817**	x0.488	x1.19	x0.853	x0.871	**x0.864**
summary: run all-flows, single hel.	x0.366	x0.860	x0.569	x0.567	**x0.595**	x0.394	x0.861	x0.454	x0.507	**x0.704**	x0.278	x1.02	x0.467	x0.541	**x0.807**

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol LC (block 3 of 3)

ID	base process	metric	n=7			n=8			n=9		
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	9.93	1.45 (r) x2.73	2.22e+03 (c)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	90.9	839 ([x0.720] x0.749)	([x0.928] x0.928)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.28	1.50 (r) x2.99	1.53e+03 (c)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	77.2	941 ([x0.737] x0.761)	([x0.810] x0.810)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	11.1	1.01 (r) x1.24	288 (c)	13.9	11.2 (r) x1.84	130 (r)	20.0	138 (r) x5.23	3.79e+03 (r)
		run [μs/pt]	31.3	105 ([x0.710] x0.751)	([x0.699] x0.700)	82.7	888 ([x0.766] x0.793)	([x0.771] x0.780)	206	1.66e+04 ([x0.729] x0.753)	([x1.60] x1.60)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	10.5	0.961 (r) x1.13	110 (c)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	19.0	96.2 ([x0.747] x0.784)	([x0.703] x0.703)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	20.4	1.10 (r) x1.32	510 (c)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	111	192 ([x0.752] x0.782)	([x0.577] x0.577)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	1.17e+03	5.67 (r) x0.725	822 (r)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	153	1.05e+04 ([x0.898] x0.942)	([x1.18] x1.19)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	60.7	1.60 (r) x2.08	2.14e+03 (c)	372	28.8 (r) x1.03	4.69e+03 (r)	3.78e+03	N/A N/A	N/A
		run [μs/pt]	65.1	1.51e+03 ([x0.944] x0.994)	([x0.619] x0.620)	171	2.61e+04 ([x0.945] x1.01)	([x2.73] x2.73)	429	N/A N/A	N/A
8	$gg \to gg + (n-2)g$	gen. [s]	55.4	216 (r) x14.1	6.59e+03 (r)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	210	6.68e+04 ([x0.628] x0.690)	([x1.85] x1.85)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	83.0	0.981 (r) x0.462	182 (c)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	205	67.0 ([x0.797] x0.818)	([x0.358] x0.358)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	75.9	0.953 (c) x0.552	26.2 (c)	108	10.2 (r) x0.168	15.1 (r)	164	120 (r) x0.186	64.3 (r)
		run [μs/pt]	25.5	12.5 ([x0.793] x0.794)	([x0.375] x0.375)	61.4	55.2 ([x0.776] x0.811)	([x0.638] x0.689)	146	319 ([x0.791] x0.811)	([x0.748] x0.769)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	238	0.954 (r) x0.397	153 (c)	794	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	70.9	109 ([x1.13] x1.19)	([x0.651] x0.652)	187	N/A N/A	N/A	N/A	N/A N/A	N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	78.7	0.900 (c) x0.998	57.9 (c)	128	10.1 (r) x0.137	18.2 (r)	181	119 (r) x0.177	94.6 (r)
		run [μs/pt]	41.5	14.3 ([x0.519] x0.519)	([x0.372] x0.372)	152	65.6 ([x0.301] x0.314)	([x0.588] x0.624)	213	373 ([x0.626] x0.534)	([x0.708] x0.733)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	58.1	0.963 (r) x3.10	643 (c)	143	14.7 (r) x0.883	1.19e+03 (r)	360	149 (r) x2.67	N/A
		run [μs/pt]	8.25	362 ([x1.32] x2.70)	([x0.478] x0.478)	21.3	4.15e+03 ([x1.88] x2.37)	([x3.35] x3.40)	54.4	5.80e+04 ([x3.57] x4.04)	N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	STATIC N/A	STATIC N/A 100 (r)	464 (c)	STATIC N/A	STATIC N/A 592 (r)	287 (r)	STATIC N/A	STATIC N/A (r) ERROR	(r) BLOCKED DEP.
		run [μs/pt]	STATIC N/A	STATIC N/A 10.6	62.8	STATIC N/A	STATIC N/A 99.0	2.49e+03	STATIC N/A	STATIC N/A ERROR	BLOCKED DEP.
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	43.7	1.02 (r) x3.62	1.12e+03 (c)	95.7	15.5 (r) x1.50	3.45e+03 (r)	231	165 (r) x4.12	(r) ERROR
		run [μs/pt]	9.93	624 ([x1.18] x1.77)	([x0.533] x0.533)	27.2	1.29e+04 ([x2.13] x2.78)	([x3.93] x3.97)	77.1	1.49e+05 ([x2.50] x2.84)	ERROR

	n=7 [r:12|c:2|e:0]					n=8 [r:6|c:0|e:0]					n=9 [r:5|c:0|e:0]				
summary: generation	x0.397	x14.1	x1.28	x2.53	**x1.27**	x0.137	x1.84	x0.956	x0.925	**x0.829**	x0.177	x5.23	x2.67	x2.48	**x2.18**
summary: run single-flow, hel. sum	x0.519	x2.70	x0.789	x1.02	**x0.839**	x0.314	x2.78	x0.910	x1.35	**x0.896**	x0.534	x4.04	x0.811	x1.80	**x1.19**
summary: run all-flows, single hel.	x0.358	x1.85	x0.636	x0.725	**x1.70**	x0.624	x3.97	x1.75	x2.03	**x3.11**	x0.733	x1.60	x0.769	x1.03	**x1.56**

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

5.2 Built-in SM best measured mode versus AmpliCol: NLC

Built-in SM best measured mode versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.10 (r) x1.16	7.24 (c) x0.614	7.21 (c) x0.737
		run [μs/pt]	0.228 ([x0.742] x1.17)	0.668 ([x0.947] x0.947)	3.57 ([x0.566] x0.567)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.01 (r) x1.20	7.77 (r) x1.07	7.40 (c) x0.733
		run [μs/pt]	0.138 ([x0.908] x1.40)	0.376 ([x1.87] x1.30)	1.93 ([x0.818] x0.818)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 (r) x0.744	9.31 (c) x0.594
		run [μs/pt]		0.466 ([x0.531] x0.732)	1.11 ([x0.596] x0.597)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	8.95 (r) x0.946	9.13 (r) x0.961
		run [μs/pt]		0.246 ([x0.361] x0.690)	0.505 ([x0.377] x0.594)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.95 (c) x0.454	10.1 (c) x0.499
		run [μs/pt]		0.936 ([x0.838] x0.839)	2.85 ([x0.631] x0.631)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.99 (c) x0.490	17.0 (c) x0.463
		run [μs/pt]		1.71 ([x0.592] x0.593)	16.5 ([x0.536] x0.536)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.35 (r) x0.950	14.0 (e) x0.330
		run [μs/pt]		1.17 ([x0.341] x0.606)	6.25 ([x0.567] x0.568)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 (c) x0.396	13.2 (c) x0.554
		run [μs/pt]		7.93 ([x0.333] x0.333)	120 ([x0.426] x0.426)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.4 (c) x0.305
		run [μs/pt]			5.70 ([x0.676] x0.676)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
			n=1 [r:2|c:0|e:0]	**n=2** [r:4|c:4|e:0]	**n=3** [r:1|c:7|e:1]
summary: generation			x1.16 | x1.20 | x1.18 | x1.18 | $\boxed{\text{x1.18}}$	x0.396 | x1.07 | x0.679 | x0.708 | $\boxed{\text{x0.695}}$	x0.305 | x0.961 | x0.554 | x0.575 | $\boxed{\text{x0.523}}$
summary: wall			x1.17 | x1.40 | x1.28 | x1.28 | $\boxed{\text{x1.26}}$	x0.333 | x1.30 | x0.711 | x0.755 | $\boxed{\text{x0.502}}$	x0.426 | x0.818 | x0.594 | x0.601 | $\boxed{\text{x0.465}}$

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol NLC (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.19	(c) x1.42	8.34	(e) x3.63	NOT RUN	
		run [μs/pt]	31.4	([x0.632] x0.633)	366	([x0.684] x0.684)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.70	(c) x1.35	7.64	(r) x4.75	NOT RUN	
		run [μs/pt]	29.9	([x0.456] x0.456)	250	([x0.416] x0.609)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.43	(c) x0.762	10.3	(c) x1.30	NOT RUN	
		run [μs/pt]	5.53	([x0.328] x0.328)	44.7	([x0.361] x0.361)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.44	(r) x0.953	9.51	(c) x0.804	NOT RUN	
		run [μs/pt]	2.50	([x0.288] x0.399)	22.3	([x0.324] x0.324)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.6	(c) x0.760	11.7	(c) x2.63	NOT RUN	
		run [μs/pt]	15.8	([x0.564] x0.564)	117	([x0.806] x0.806)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	33.1	(e) x0.616	63.4	(e) x11.4	NOT RUN	
		run [μs/pt]	230	([x0.543] x0.544)	3.46e+03	([x0.710] x0.710)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	18.7	(r) x0.695	28.1	(e) x1.39	NOT RUN	
		run [μs/pt]	57.7	([x0.356] x0.522)	695	([x0.599] x0.599)		
8	$gg \to gg + (n-2)g$	gen. [s]	17.0	(r) x1.17	22.1	(r) x2.80	NOT RUN	
		run [μs/pt]	2.01e+03	([x0.355] x0.546)	3.66e+04	([x0.355] x0.587)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.2	(c) x0.403	26.7	(r) x0.704	NOT RUN	
		run [μs/pt]	15.0	([x0.739] x0.740)	74.7	([x0.700] x0.797)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	58.9	(c) x0.121	63.6	(c) x0.134	NOT RUN	
		run [μs/pt]	3.91	([x0.496] x0.497)	8.84	([x0.428] x0.428)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.7	(r) x0.285	69.2	(r) x0.191	NOT RUN	
		run [μs/pt]	11.8	([x0.584] x0.645)	65.1	([x0.563] x0.616)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6	(c) x0.196	72.0	(c) x0.219	NOT RUN	
		run [μs/pt]	7.35	([x0.284] x0.284)	19.4	([x0.206] x0.206)		
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	0.00152	(r) x6.33e+03	0.0101	(e) x860	0.101	(r) x657
		run [μs/pt]	144	([x0.0233] x0.0249)	1.28e+03	([x0.0240] x0.0240)	1.49e+04	([x0.0223] x0.0288)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		STATIC N/A	31.0 (r)
		run [μs/pt]					STATIC N/A	76.6
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	0.00198	(e) x2.52e+03	0.0112	(e) x1.06e+03	0.108	(e) x922
		run [μs/pt]	219	([x0.0254] x0.0254)	2.02e+03	([x0.0241] x0.0241)	2.40e+04	([x0.0342] x0.0342)

	n=4 [r:5|c:7|e:2]	n=5 [r:4|c:5|e:5]	n=6 [r:1|c:0|e:1]
summary: generation	x0.121 | x6.33e+03 | x0.761 | x633 | **x0.520**	x0.134 | x1.06e+03 | x2.01 | x139 | **x2.60**	x657 | x922 | x790 | x790 | **x794**
summary: wall	x0.0249 | x0.740 | x0.509 | x0.443 | **x0.477**	x0.0240 | x0.806 | x0.593 | x0.484 | **x0.557**	x0.0288 | x0.0342 | x0.0315 | x0.0315 | **x0.0321**

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

5.3 Built-in SM best measured mode versus AmpliCol: full-colour

Built-in SM best measured mode versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.17 (r) x1.13	7.21 (c) x0.596	7.26 (c) x0.742
		run [μs/pt]	0.270 ([x0.619] x0.981)	0.744 ([x0.852] x0.852)	3.97 ([x0.504] x0.504)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.45 (r) x1.10	7.59 (r) x1.08	7.90 (r) x1.20
		run [μs/pt]	0.159 ([x0.801] x1.23)	0.428 ([x0.810] x1.11)	2.14 ([x0.620] x0.854)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	9.28 (r) x0.962	9.55 (c) x0.635
		run [μs/pt]		0.522 ([x0.658] x0.650)	1.20 ([x0.543] x0.544)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	163 (r) x0.0559	9.24 (r) x0.992
		run [μs/pt]		0.365 ([x0.249] x0.446)	0.559 ([x0.476] x0.666)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.91 (c) x0.438	10.2 (c) x0.502
		run [μs/pt]		1.05 ([x0.717] x0.717)	3.08 ([x0.588] x0.588)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.91 (c) x0.517	16.5 (c) x0.476
		run [μs/pt]		1.98 ([x0.535] x0.535)	19.5 ([x0.469] x0.470)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.57 (r) x0.890	13.7 (e) x0.336
		run [μs/pt]		1.48 ([x0.291] x0.484)	7.52 ([x0.469] x0.469)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.8 (c) x0.369	13.6 (c) x0.547
		run [μs/pt]		9.31 ([x0.275] x0.275)	146 ([x0.353] x0.354)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.3 (c) x0.318
		run [μs/pt]			6.11 ([x0.640] x0.641)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1 [r:2|c:0|e:0]					n=2 [r:4|c:4|e:0]					n=3 [r:2|c:6|e:1]				
summary: generation	x1.10	x1.13	x1.12	x1.12	**x1.12**	x0.0559	x1.08	x0.557	x0.614	**x0.230**	x0.318	x1.20	x0.547	x0.638	**x0.571**
summary: wall	x0.981	x1.23	x1.11	x1.11	**x1.07**	x0.275	x1.11	x0.593	x0.634	**x0.422**	x0.354	x0.854	x0.544	x0.565	**x0.394**

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.15	(c) x1.49	8.47	(e) x3.65	NOT RUN	
		run [μs/pt]	36.1	([x0.554] x0.554)	464	([x0.570] x0.571)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.75	(c) x1.38	8.32	(r) x4.00	NOT RUN	
		run [μs/pt]	20.6	([x0.544] x0.544)	283	([x0.371] x0.534)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.59	(c) x0.712	9.87	(c) x1.35	NOT RUN	
		run [μs/pt]	6.11	([x0.297] x0.298)	51.6	([x0.316] x0.316)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.27	(r) x0.965	9.73	(c) x0.793	NOT RUN	
		run [μs/pt]	2.80	([x0.259] x0.359)	25.4	([x0.287] x0.288)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	(c) x0.811	11.5	(c) x2.71	NOT RUN	
		run [μs/pt]	16.7	([x0.545] x0.546)	131	([x0.728] x0.728)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	31.2	(e) x0.647	63.2	(e) x16.6	NOT RUN	
		run [μs/pt]	283	([x0.480] x0.480)	6.16e+03	([x0.510] x0.511)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	19.3	(r) x0.692	28.0	(e) x1.41	NOT RUN	
		run [μs/pt]	74.3	([x0.279] x0.392)	1.09e+03	([x0.419] x0.419)		
8	$gg \to gg + (n-2)g$	gen. [s]	16.3	(r) x1.24	22.2	(c) x7.55	NOT RUN	
		run [μs/pt]	3.06e+03	([x0.227] x0.415)	9.67e+04	([x0.307] x0.307)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.5	(c) x0.398	26.8	(r) x0.699	NOT RUN	
		run [μs/pt]	15.8	([x0.706] x0.707)	79.5	([x0.653] x0.743)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	59.5	(c) x0.121	63.8	(c) x0.138	NOT RUN	
		run [μs/pt]	4.13	([x0.468] x0.468)	9.25	([x0.412] x0.412)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.5	(e) x0.152	69.4	(r) x0.195	NOT RUN	
		run [μs/pt]	12.8	([x0.605] x0.605)	68.6	([x0.533] x0.575)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6	(c) x0.191	71.9	(c) x0.229	NOT RUN	
		run [μs/pt]	7.59	([x0.287] x0.287)	20.0	([x0.203] x0.203)		
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	0.00151	(r) x6.27e+03	0.0101	(e) x933	0.104	(r) x654
		run [μs/pt]	150	([x0.0162] x0.0234)	1.31e+03	([x0.0233] x0.0233)	1.64e+04	([x0.0203] x0.0265)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		STATIC N/A	30.8 (r)
		run [μs/pt]					STATIC N/A	77.2
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	0.00198	(c) x8.25e+03	0.0112	(e) x1.04e+03	0.110	(e) x920
		run [μs/pt]	225	([x0.0297] x0.0297)	2.07e+03	([x0.0237] x0.0237)	2.59e+04	([x0.0320] x0.0320)

	n=4 [r:4|c:8|e:2]	n=5 [r:3|c:6|e:5]	n=6 [r:1|c:0|e:1]
summary: generation	x0.121 | x8.25e+03 | x0.761 | x1.04e+03 | **x0.549**	x0.138 | x1.04e+03 | x2.06 | x144 | **x3.69**	x654 | x920 | x787 | x787 | **x790**
summary: wall	x0.0234 | x0.707 | x0.442 | x0.408 | **x0.386**	x0.0233 | x0.743 | x0.416 | x0.404 | **x0.313**	x0.0265 | x0.0320 | x0.0292 | x0.0292 | **x0.0299**

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μs/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

The detailed tables are grouped by execution mode, model source, and colour treatment.

5.4 Built-in SM recurrence versus AmpliCol LC

Built-in SM recurrence versus AmpliCol LC (block 1 of 3)

ID	base process	metric	n=1				n=2				n=3			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.14	0.000973	x1.16	8.12	7.15	0.00107	x1.25	8.21	7.32	0.000760	x1.20	8.20
		run [μs/pt]	0.0561	0.122	([x3.05] x4.36)	([x0.185] x1.17)	0.362	0.280	([x1.47] x1.83)	([x0.377] x1.07)	1.23	0.770	([x1.10] x1.32)	([x0.528] x0.916)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.72	0.000448	x1.04	8.05	7.89	0.000850	x1.07	9.16	7.61	0.000766	x1.29	9.32
		run [μs/pt]	0.0323	0.124	([x3.82] x5.81)	([x0.171] x1.10)	0.219	0.284	([x1.66] x2.23)	([x0.366] x1.05)	0.752	0.748	([x1.30] x1.57)	([x0.543] x0.937)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE				9.36	0.00106	x0.872	8.27	9.52	0.00149	x0.927	8.25
		run [μs/pt]					0.185	0.268	([x1.37] x1.88)	([x0.384] x0.904)	0.453	0.430	([x1.16] x1.51)	([x0.507] x0.993)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE				9.18	0.000491	x0.930	8.47	9.31	0.000688	x0.892	8.75
		run [μs/pt]					0.0665	0.212	([x1.50] x2.77)	([x0.343] x1.01)	0.139	0.363	([x1.39] x2.40)	([x0.417] x0.999)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE				9.90	0.000496	x0.836	8.40	10.1	0.000726	x0.906	8.74
		run [μs/pt]					0.469	0.295	([x1.66] x2.00)	([x0.356] x1.04)	2.07	0.605	([x1.08] x1.20)	([x0.582] x1.08)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.90	0.000518	x0.995	9.16	17.0	0.000889	x0.640	8.31
		run [μs/pt]					0.319	0.416	([x2.14] x2.64)	([x0.497] x1.05)	1.95	1.62	([x1.56] x1.66)	([x0.669] x0.892)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				9.22	0.00106	x0.892	8.96	13.3	0.000754	x0.699	8.27
		run [μs/pt]					0.146	0.283	([x1.96] x2.76)	([x0.289] x0.905)	0.625	0.918	([x1.37] x1.76)	([x0.508] x0.808)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE				11.1	0.000548	x0.805	8.22	13.1	0.00217	x0.822	9.13
		run [μs/pt]					0.459	0.596	([x1.30] x1.58)	([x0.976] x1.45)	2.39	3.67	([x1.23] x1.35)	([x1.21] x1.36)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				19.3	0.00147	x0.506	8.74
		run [μs/pt]									4.57	0.849	([x1.07] x1.16)	([x0.483] x0.896)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												

	n=1					n=2					n=3				
summary: generation	x1.04	x1.16	x1.10	x1.10	$\boxed{x1.10}$	x0.805	x1.25	x0.911	x0.956	$\boxed{x0.940}$	x0.506	x1.29	x0.892	x0.875	$\boxed{x0.803}$
summary: run single-flow, hel. sum	x4.36	x5.81	x5.09	x5.09	$\boxed{x4.89}$	x1.58	x2.77	x2.12	x2.21	$\boxed{x2.07}$	x1.16	x2.40	x1.51	x1.55	$\boxed{x1.35}$
summary: run all-flows, single hel.	x1.10	x1.17	x1.13	x1.13	$\boxed{x1.13}$	x0.904	x1.45	x1.04	x1.06	$\boxed{x1.11}$	x0.808	x1.36	x0.937	x0.987	$\boxed{x1.08}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol LC (block 2 of 3)

ID	base process	metric	n=4				n=5				n=6			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.84	0.00489	x1.23	8.33	8.16	0.0149	x1.28	9.44	8.78	0.140	x2.33	17.9
		run [μs/pt]	4.46	3.05	([x1.01] x1.11)	([x0.646] x0.802)	13.2	15.3	([x0.758] x0.820)	([x0.647] x0.706)	34.1	96.3	([x0.753] x1.48)	([x0.661] x0.685)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.83	0.00238	x1.30	9.83	8.15	0.0149	x1.26	10.7	8.40	0.149	x1.60	19.2
		run [μs/pt]	2.52	4.21	([x1.16] x1.61)	([x0.482] x0.600)	9.00	15.4	([x0.989] x1.06)	([x0.670] x0.732)	26.6	98.6	([x0.651] x0.689)	([x0.687] x0.706)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.42	0.00201	x0.943	8.30	9.89	0.0115	x0.921	8.37	11.2	0.0978	x0.942	10.3
		run [μs/pt]	1.24	0.964	([x1.02] x1.22)	([x0.582] x0.913)	3.85	3.70	([x1.12] x1.20)	([x0.633] x0.775)	11.1	16.7	([x0.799] x0.876)	([x0.677] x0.743)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.29	0.00197	x0.955	8.33	9.41	0.0114	x0.917	9.08	10.1	0.0971	x0.935	9.55
		run [μs/pt]	0.413	0.831	([x1.19] x1.76)	([x0.569] x0.959)	1.66	3.15	([x1.21] x1.41)	([x0.905] x1.09)	6.05	15.2	([x1.19] x1.31)	([x0.673] x0.740)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	0.00433	x0.899	8.22	11.5	0.0123	x0.969	8.79	13.4	0.102	x1.17	12.0
		run [μs/pt]	6.36	1.68	([x0.932] x1.02)	([x0.548] x0.825)	16.7	6.30	([x0.864] x0.932)	([x0.685] x0.814)	42.9	30.6	([x0.804] x0.846)	([x0.644] x0.695)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	30.8	0.00328	x0.676	9.07	64.5	0.0348	x1.15	13.7	186	0.296	x1.30	59.9
		run [μs/pt]	6.89	8.41	([x0.964] x1.39)	([x0.874] x0.951)	21.2	77.5	([x1.09] x0.888)	([x0.478] x0.861)	58.0	479	([x0.932] x1.07)	([x0.969] x1.02)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	17.6	0.00499	x0.611	8.93	23.6	0.0146	x0.732	9.93	33.2	0.137	x1.31	22.7
		run [μs/pt]	2.39	4.16	([x1.43] x1.62)	([x0.655] x0.772)	8.08	23.6	([x0.946] x1.07)	([x0.576] x0.612)	23.6	157	([x0.859] x0.934)	([x0.669] x0.683)
8	$gg \to gg + (n-2)g$	gen. [s]	15.8	0.00514	x1.15	11.4	18.3	0.0838	x2.87	37.3	27.8	3.09	x4.83	319
		run [μs/pt]	9.36	26.2	([x0.712] x0.760)	([x0.727] x0.776)	28.7	221	([x0.650] x0.682)	([x0.757] x0.775)	79.0	2.74e+03	([x0.638] x0.666)	([x1.08] x1.09)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.3	0.00206	x0.496	8.36	26.8	0.0112	x0.518	8.58	38.4	0.0927	x0.557	10.8
		run [μs/pt]	12.9	1.56	([x0.932] x1.00)	([x0.573] x0.894)	33.3	3.94	([x0.848] x0.894)	([x0.643] x0.844)	84.1	14.5	([x0.806] x0.841)	([x0.648] x0.753)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	57.0	0.00195	x0.164	9.17	61.7	0.0111	x0.146	8.32	65.9	0.0922	x0.159	8.83
		run [μs/pt]	2.06	1.29	([x1.27] x1.45)	([x0.562] x0.830)	4.66	1.92	([x1.03] x1.15)	([x0.576] x0.833)	10.7	3.90	([x0.900] x0.981)	([x0.649] x0.844)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	31.1	0.00201	x0.299	8.77	64.4	0.0104	x0.232	8.95	117	0.107	x0.259	9.97
		run [μs/pt]	2.99	1.63	([x1.39] x1.55)	([x0.634] x0.875)	9.01	4.83	([x1.22] x1.32)	([x0.682] x0.827)	26.0	36.5	([x1.12] x1.19)	([x0.358] x0.395)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	63.3	0.00432	x0.145	8.90	67.5	0.0110	x0.142	8.39	65.6	0.0913	x0.154	8.82
		run [μs/pt]	3.81	1.85	([x0.719] x0.817)	([x0.602] x0.779)	7.86	2.74	([x0.634] x0.720)	([x0.552] x0.720)	17.8	4.96	([x0.562] x0.614)	([x0.625] x0.776)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	12.4	0.00149	x0.745	8.64	21.5	0.00985	x0.579	9.72	31.6	0.0946	x1.38	14.0
		run [μs/pt]	0.457	2.07	([x1.36] x2.32)	([x0.732] x0.929)	1.14	9.33	([x1.19] x2.08)	([x0.755] x0.846)	3.05	50.9	([x1.15] x1.91)	([x0.875] x0.908)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				STATIC N/A	STATIC N/A	17.4	11.8
		run [μs/pt]									STATIC N/A	STATIC N/A	7.05	22.0
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	11.9	0.00195	x0.789	9.18	18.7	0.0110	x0.653	10.4	26.4	0.101	x1.54	17.9
		run [μs/pt]	0.678	3.22	([x1.16] x2.61)	([x0.837] x0.995)	1.58	15.8	([x1.11] x1.92)	([x0.794] x0.856)	3.90	87.2	([x1.03] x1.77)	([x1.03] x1.06)

	n=4					n=5					n=6				
summary: generation	x0.145	x1.30	x0.767	x0.742	$\boxed{\text{x0.502}}$	x0.142	x2.87	x0.825	x0.884	$\boxed{\text{x0.642}}$	x0.154	x4.83	x1.23	x1.32	$\boxed{\text{x1.00}}$
summary: run single-flow, hel. sum	x0.760	x2.61	x1.42	x1.45	$\boxed{\text{x1.15}}$	x0.682	x2.08	x1.07	x1.15	$\boxed{\text{x0.926}}$	x0.614	x1.91	x0.957	x1.08	$\boxed{\text{x0.926}}$
summary: run all-flows, single hel.	x0.600	x0.995	x0.852	x0.850	$\boxed{\text{x0.818}}$	x0.612	x1.09	x0.820	x0.807	$\boxed{\text{x0.787}}$	x0.395	x1.09	x0.748	x0.792	$\boxed{\text{x1.03}}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol LC (block 3 of 3)

ID	base process	metric	n=7				n=8				n=9			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	9.93	1.45	x2.73	113	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	90.9	839	([x0.720] x0.749)	([x0.947] x0.957)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.28	1.50	x2.99	123	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	77.2	941	([x0.737] x0.761)	([x1.20] x0.945)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	11.1	1.01	x1.24	19.4	13.9	11.2	x1.84	130	20.0	138	x5.23	3.79e+03
		run [μs/pt]	31.3	105	([x0.710] x0.751)	([x0.678] x0.708)	82.7	888	([x0.766] x0.793)	([x0.771] x0.780)	206	1.66e+04	([x0.729] x0.753)	([x1.60] x1.60)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	10.5	0.961	x1.13	19.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	19.0	96.2	([x0.747] x0.784)	([x0.693] x0.723)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	20.4	1.10	x1.32	36.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	111	192	([x0.752] x0.782)	([x0.687] x0.737)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	1.17e+03	5.67	x0.725	822	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	153	1.05e+04	([x0.898] x0.942)	([x1.18] x1.19)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	60.7	1.60	x2.08	178	372	28.8	x1.03	4.69e+03	3.78e+03	N/A	N/A	N/A
		run [μs/pt]	65.1	1.51e+03	([x0.944] x0.994)	([x1.23] x1.23)	171	2.61e+04	([x0.945] x1.01)	([x2.73] x2.73)	429	N/A	N/A	N/A
8	$gg \to gg + (n-2)g$	gen. [s]	55.4	216	x14.1	6.59e+03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	210	6.68e+04	([x0.628] x0.690)	([x1.85] x1.85)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	83.0	0.981	x0.462	21.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	205	67.0	([x0.797] x0.818)	([x0.608] x0.652)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	75.9	0.953	x0.174	9.59	108	10.2	x0.168	15.1	164	120	x0.186	64.3
		run [μs/pt]	25.5	12.5	([x0.869] x0.928)	([x0.688] x0.791)	61.4	55.2	([x0.776] x0.811)	([x0.638] x0.689)	146	319	([x0.791] x0.811)	([x0.748] x0.769)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	238	0.954	x0.397	18.0	794	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	70.9	109	([x1.13] x1.19)	([x0.668] x0.693)	187	N/A	N/A	N/A	N/A	N/A	N/A	N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	78.7	0.900	x0.158	10.6	128	10.1	x0.137	18.2	181	119	x0.177	94.6
		run [μs/pt]	41.5	14.3	([x0.517] x0.557)	([x0.711] x0.805)	152	65.6	([x0.301] x0.314)	([x0.588] x0.624)	213	373	([x0.626] x0.534)	([x0.708] x0.733)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	58.1	0.963	x3.10	59.8	143	14.7	x0.883	1.19e+03	360	149	x2.67	N/A
		run [μs/pt]	8.25	362	([x1.32] x2.70)	([x1.30] x1.31)	21.3	4.15e+03	([x1.88] x2.37)	([x3.35] x3.40)	54.4	5.80e+04	([x3.57] x4.04)	N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	STATIC N/A	STATIC N/A	100	29.8	STATIC N/A	STATIC N/A	592	287	STATIC N/A	STATIC N/A	ERROR	BLOCKED DEP.
		run [μs/pt]	STATIC N/A	STATIC N/A	10.6	175	STATIC N/A	STATIC N/A	99.0	2.49e+03	STATIC N/A	STATIC N/A	ERROR	BLOCKED DEP.
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	43.7	1.02	x3.62	119	95.7	15.5	x1.50	3.45e+03	231	165	x4.12	ERROR
		run [μs/pt]	9.93	624	([x1.18] x1.77)	([x2.24] x2.25)	27.2	1.29e+04	([x2.13] x2.78)	([x3.93] x3.97)	77.1	1.49e+05	([x2.50] x2.84)	ERROR

	n=7					n=8					n=9				
summary: generation	x0.158	x14.1	x1.28	x2.44	**x1.22**	x0.137	x1.84	x0.956	x0.925	**x0.829**	x0.177	x5.23	x2.67	x2.48	**x2.18**
summary: run single-flow, hel. sum	x0.557	x2.70	x0.801	x1.03	**x0.844**	x0.314	x2.78	x0.910	x1.35	**x0.896**	x0.534	x4.04	x0.811	x1.80	**x1.19**
summary: run all-flows, single hel.	x0.652	x2.25	x0.875	x1.06	**x1.73**	x0.624	x3.97	x1.75	x2.03	**x3.11**	x0.733	x1.60	x0.769	x1.03	**x1.56**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.5 Built-in SM recurrence versus AmpliCol NLC

Built-in SM recurrence versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.10 x1.16	7.24 x1.14	7.21 x1.27
		run [μs/pt]	0.228 ([x0.742] x1.17)	0.668 ([x0.821] x1.10)	3.57 ([x0.663] x0.905)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.01 x1.20	7.77 x1.07	7.40 x1.22
		run [μs/pt]	0.138 ([x0.908] x1.40)	0.376 ([x1.87] x1.30)	1.93 ([x0.687] x0.946)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 x0.744	9.31 x0.889
		run [μs/pt]		0.466 ([x0.531] x0.732)	1.11 ([x0.515] x0.659)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	8.95 x0.946	9.13 x0.961
		run [μs/pt]		0.246 ([x0.361] x0.690)	0.505 ([x0.377] x0.594)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.95 x0.891	10.1 x0.904
		run [μs/pt]		0.936 ([x0.810] x1.08)	2.85 ([x0.797] x0.972)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.99 x1.07	17.0 x0.613
		run [μs/pt]		1.71 ([x0.641] x0.986)	16.5 ([x0.556] x0.867)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.35 x0.950	14.0 x0.629
		run [μs/pt]		1.17 ([x0.341] x0.606)	6.25 ([x0.389] x0.624)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 x0.835	13.2 x0.876
		run [μs/pt]		7.93 ([x0.436] x0.632)	120 ([x0.351] x0.466)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.4 x0.478
		run [μs/pt]			5.70 ([x0.944] x1.08)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1					n=2					n=3				
summary: generation	x1.16	x1.20	x1.18	x1.18	**x1.18**	x0.744	x1.14	x0.948	x0.957	**x0.941**	x0.478	x1.27	x0.889	x0.871	**x0.791**
summary: wall	x1.17	x1.40	x1.28	x1.28	**x1.26**	x0.606	x1.30	x0.859	x0.891	**x0.752**	x0.466	x1.08	x0.867	x0.791	**x0.562**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol NLC (block 2 of 2)

ID	base process	metric	n=4	n=5	n=6
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.19 x1.50	8.34 x5.77	NOT RUN
		run [μs/pt]	31.4 ([x0.516] x0.754)	366 ([x0.529] x0.779)	
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.70 x1.87	7.64 x4.75	NOT RUN
		run [μs/pt]	29.9 ([x0.326] x0.488)	250 ([x0.416] x0.609)	
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.43 x0.912	10.3 x1.19	NOT RUN
		run [μs/pt]	5.53 ([x0.398] x0.513)	44.7 ([x0.318] x0.432)	
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.44 x0.953	9.51 x1.05	NOT RUN
		run [μs/pt]	2.50 ([x0.288] x0.399)	22.3 ([x0.306] x0.376)	
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.6 x1.01	11.7 x2.16	NOT RUN
		run [μs/pt]	15.8 ([x0.646] x0.803)	117 ([x0.659] x0.856)	
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	33.1 x0.842	63.4 x4.21	NOT RUN
		run [μs/pt]	230 ([x0.476] x0.720)	3.46e+03 ([x2.04] x2.66)	
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	18.7 x0.695	28.1 x1.89	NOT RUN
		run [μs/pt]	57.7 ([x0.356] x0.522)	695 ([x0.523] x0.788)	
8	$gg \to gg + (n-2)g$	gen. [s]	17.0 x1.17	22.1 x2.80	NOT RUN
		run [μs/pt]	2.01e+03 ([x0.355] x0.546)	3.66e+04 ([x0.355] x0.587)	
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.2 x0.527	26.7 x0.704	NOT RUN
		run [μs/pt]	15.0 ([x0.790] x0.892)	74.7 ([x0.700] x0.797)	
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	58.9 x0.156	63.6 x0.149	NOT RUN
		run [μs/pt]	3.91 ([x0.687] x0.758)	8.84 ([x0.544] x0.600)	
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.7 x0.285	69.2 x0.191	NOT RUN
		run [μs/pt]	11.8 ([x0.584] x0.645)	65.1 ([x0.563] x0.616)	
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6 x0.131	72.0 x0.125	NOT RUN
		run [μs/pt]	7.35 ([x0.376] x0.427)	19.4 ([x0.264] x0.294)	
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	0.00152 x6.33e+03	0.0101 x1.42e+03	0.101 x657
		run [μs/pt]	144 ([x0.0233] x0.0249)	1.28e+03 ([x0.0155] x0.0304)	1.49e+04 ([x0.0223] x0.0288)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	STATIC N/A 31.0
		run [μs/pt]			STATIC N/A 76.6
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	0.00198 x4.78e+03	0.0112 x1.84e+03	0.108 x1.14e+03
		run [μs/pt]	219 ([x0.0254] x0.0278)	2.02e+03 ([x0.0184] x0.0259)	2.40e+04 ([x0.0375] x0.0439)

	n=4	n=5	n=6
summary: generation	x0.131 | x6.33e+03 | x0.932 | x794 | $\boxed{\text{x0.592}}$	x0.125 | x1.84e+03 | x2.02 | x234 | $\boxed{\text{x1.53}}$	x657 | x1.14e+03 | x899 | x899 | $\boxed{\text{x907}}$
summary: wall	x0.0249 | x0.892 | x0.534 | x0.537 | $\boxed{\text{x0.497}}$	x0.0259 | x2.66 | x0.605 | x0.675 | $\boxed{\text{x0.710}}$	x0.0288 | x0.0439 | x0.0364 | x0.0364 | $\boxed{\text{x0.0381}}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.6 Built-in SM recurrence versus AmpliCol full-colour

Built-in SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.17 x1.13	7.21 x1.14	7.26 x1.24
		run [μs/pt]	0.270 ([x0.619] x0.981)	0.744 ([x0.701] x0.962)	3.97 ([x0.623] x0.847)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.45 x1.10	7.59 x1.08	7.90 x1.20
		run [μs/pt]	0.159 ([x0.801] x1.23)	0.428 ([x0.810] x1.11)	2.14 ([x0.620] x0.854)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	9.28 x0.962	9.55 x0.954
		run [μs/pt]		0.522 ([x0.658] x0.650)	1.20 ([x0.476] x0.606)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	163 x0.0559	9.24 x0.992
		run [μs/pt]		0.365 ([x0.249] x0.446)	0.559 ([x0.476] x0.666)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.91 x0.838	10.2 x0.902
		run [μs/pt]		1.05 ([x0.791] x1.03)	3.08 ([x0.710] x0.867)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.91 x1.02	16.5 x0.645
		run [μs/pt]		1.98 ([x0.612] x0.905)	19.5 ([x0.522] x0.882)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.57 x0.890	13.7 x0.699
		run [μs/pt]		1.48 ([x0.291] x0.484)	7.52 ([x0.339] x0.519)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.8 x0.778	13.6 x0.837
		run [μs/pt]		9.31 ([x0.387] x0.557)	146 ([x0.292] x0.395)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.3 x0.499
		run [μs/pt]			6.11 ([x0.807] x0.938)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1					n=2					n=3				
summary: generation	x1.10	x1.13	x1.12	x1.12	**x1.12**	x0.0559	x1.14	x0.926	x0.845	**x0.306**	x0.499	x1.24	x0.902	x0.885	**x0.813**
summary: wall	x0.981	x1.23	x1.11	x1.11	**x1.07**	x0.446	x1.11	x0.778	x0.768	**x0.660**	x0.395	x0.938	x0.847	x0.730	**x0.491**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.15	x1.55	8.47	x5.71	NOT RUN	
		run [μs/pt]	36.1	([x0.451] x0.738)	464	([x0.416] x0.613)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.75	x1.39	8.32	x4.00	NOT RUN	
		run [μs/pt]	20.6	([x0.460] x0.719)	283	([x0.371] x0.534)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.59	x0.918	9.87	x1.19	NOT RUN	
		run [μs/pt]	6.11	([x0.346] x0.446)	51.6	([x0.274] x0.411)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.27	x0.965	9.73	x0.973	NOT RUN	
		run [μs/pt]	2.80	([x0.259] x0.359)	25.4	([x0.271] x0.351)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	x0.995	11.5	x2.12	NOT RUN	
		run [μs/pt]	16.7	([x0.631] x0.784)	131	([x0.579] x0.821)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	31.2	x0.955	63.2	x4.24	NOT RUN	
		run [μs/pt]	283	([x0.394] x0.662)	6.16e+03	([x1.15] x1.63)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	19.3	x0.692	28.0	x1.90	NOT RUN	
		run [μs/pt]	74.3	([x0.279] x0.392)	1.09e+03	([x0.344] x0.550)		
8	$gg \to gg + (n-2)g$	gen. [s]	16.3	x1.24	22.2	x3.15	NOT RUN	
		run [μs/pt]	3.06e+03	([x0.227] x0.415)	9.67e+04	([x0.134] x0.372)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.5	x0.515	26.8	x0.699	NOT RUN	
		run [μs/pt]	15.8	([x0.757] x0.836)	79.5	([x0.653] x0.743)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	59.5	x0.146	63.8	x0.147	NOT RUN	
		run [μs/pt]	4.13	([x0.632] x0.702)	9.25	([x0.612] x0.668)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.5	x0.282	69.4	x0.195	NOT RUN	
		run [μs/pt]	12.8	([x0.565] x0.625)	68.6	([x0.533] x0.575)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6	x0.135	71.9	x0.134	NOT RUN	
		run [μs/pt]	7.59	([x0.367] x0.406)	20.0	([x0.254] x0.280)		
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	0.00151	x6.27e+03	0.0101	x1.42e+03	0.104	x654
		run [μs/pt]	150	([x0.0162] x0.0234)	1.31e+03	([x0.0210] x0.0233)	1.64e+04	([x0.0203] x0.0265)
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		STATIC N/A	30.8
		run [μs/pt]					STATIC N/A	77.2
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	0.00198	x5.39e+03	0.0112	x1.84e+03	0.110	x1.19e+03
		run [μs/pt]	225	([x0.0183] x0.0333)	2.07e+03	([x0.0178] x0.0252)	2.59e+04	([x0.0385] x0.0450)

	n=4					n=5					n=6				
summary: generation	x0.135	x6.27e+03	x0.960	x833	$\boxed{\text{x0.594}}$	x0.134	x1.84e+03	x2.01	x235	$\boxed{\text{x1.54}}$	x654	x1.19e+03	x921	x921	$\boxed{\text{x929}}$
summary: wall	x0.0234	x0.836	x0.536	x0.510	$\boxed{\text{x0.405}}$	x0.0233	x1.63	x0.542	x0.543	$\boxed{\text{x0.437}}$	x0.0265	x0.0450	x0.0358	x0.0358	$\boxed{\text{x0.0379}}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.7 UFO-SM recurrence versus AmpliCol LC

UFO-SM recurrence versus AmpliCol LC (block 1 of 3)

<table>
<tr><th>ID</th><th>base process</th><th>metric</th><th>n=1</th><th>n=2</th><th>n=3</th></tr>
<tr><td>1</td><td>$d\bar{d} \to Z + (n-1)g$</td><td>gen. [s]</td><td>7.14 | 0.000973 x2.13 | 15.2</td><td>7.15 | 0.00107 x2.06 | 14.4</td><td>7.32 | 0.000760 x2.13 | 14.5</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td>0.0561 | 0.122 ([x3.28] x4.60) | ([x0.200] x1.14)</td><td>0.362 | 0.280 ([x1.67] x2.04) | ([x0.387] x1.25)</td><td>1.23 | 0.770 ([x1.21] x1.44) | ([x0.666] x1.06)</td></tr>
<tr><td>2</td><td>$u\bar{d} \to W^+ + (n-1)g$</td><td>gen. [s]</td><td>7.72 | 0.000448 x1.94 | 15.3</td><td>7.89 | 0.000850 x1.99 | 14.8</td><td>7.61 | 0.000766 x2.11 | 17.1</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td>0.0323 | 0.124 ([x3.95] x5.87) | ([x0.154] x1.13)</td><td>0.219 | 0.284 ([x1.59] x2.14) | ([x0.358] x1.05)</td><td>0.752 | 0.748 ([x1.41] x1.60) | ([x0.590] x0.995)</td></tr>
<tr><td>3</td><td>$d\bar{d} \to e^+e^- + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>9.36 | 0.00106 x1.65 | 14.4</td><td>9.52 | 0.00149 x1.58 | 14.8</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.185 | 0.268 ([x1.41] x1.92) | ([x0.374] x0.899)</td><td>0.453 | 0.430 ([x1.31] x1.62) | ([x0.468] x0.952)</td></tr>
<tr><td>4</td><td>$u\bar{d} \to e^+\nu_e + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>9.18 | 0.000491 x1.59 | 14.3</td><td>9.31 | 0.000688 x1.56 | 14.9</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.0665 | 0.212 ([x1.41] x2.77) | ([x0.263] x0.940)</td><td>0.139 | 0.363 ([x1.45] x2.45) | ([x0.374] x0.960)</td></tr>
<tr><td>5</td><td>$d\bar{d} \to ZZ + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>9.90 | 0.000496 x1.50 | 14.3</td><td>10.1 | 0.000726 x1.53 | 15.1</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.469 | 0.295 ([x1.94] x2.26) | ([x0.367] x1.07)</td><td>2.07 | 0.605 ([x1.37] x1.52) | ([x0.526] x1.02)</td></tr>
<tr><td>6</td><td>$gg \to t\bar{t} + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>8.90 | 0.000518 x1.66 | 15.4</td><td>17.0 | 0.000889 x1.03 | 15.5</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.319 | 0.416 ([x2.32] x2.89) | ([x0.412] x0.959)</td><td>1.95 | 1.62 ([x1.33] x1.51) | ([x0.687] x0.916)</td></tr>
<tr><td>7</td><td>$d\bar{d} \to t\bar{t} + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>9.22 | 0.00106 x1.59 | 14.9</td><td>13.3 | 0.000754 x1.20 | 15.3</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.146 | 0.283 ([x1.96] x2.74) | ([x0.291] x0.919)</td><td>0.625 | 0.918 ([x1.44] x1.84) | ([x0.544] x0.838)</td></tr>
<tr><td>8</td><td>$gg \to gg + (n-2)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>11.1 | 0.000548 x1.32 | 14.3</td><td>13.1 | 0.00217 x1.27 | 14.8</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td>0.459 | 0.596 ([x1.43] x1.70) | ([x0.788] x1.24)</td><td>2.39 | 3.67 ([x1.27] x1.43) | ([x1.47] x1.59)</td></tr>
<tr><td>9</td><td>$d\bar{d} \to ZZZ + (n-3)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>19.3 | 0.00147 x0.800 | 14.4</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td>4.57 | 0.849 ([x1.30] x1.39) | ([x0.504] x0.913)</td></tr>
<tr><td>10</td><td>$d\bar{d} \to e^+e^-ZH + (n-4)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
<tr><td>11</td><td>$d\bar{d} \to t\bar{t}ZH + (n-4)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
<tr><td>12</td><td>$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
<tr><td>13</td><td>$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
<tr><td>14</td><td>$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
<tr><td>15</td><td>$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$</td><td>gen. [s]</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td></tr>
<tr><td></td><td></td><td>run [μs/pt]</td><td></td><td></td><td></td></tr>
</table>

<table>
<tr><th></th><th>n=1</th><th>n=2</th><th>n=3</th></tr>
<tr><td>summary: generation</td><td>x1.94 | x2.13 | x2.04 | x2.04 | x2.03</td><td>x1.32 | x2.06 | x1.62 | x1.67 | x1.64</td><td>x0.800 | x2.13 | x1.53 | x1.47 | x1.34</td></tr>
<tr><td>summary: run single-flow, hel. sum</td><td>x4.60 | x5.87 | x5.24 | x5.24 | x5.06</td><td>x1.70 | x2.89 | x2.20 | x2.31 | x2.21</td><td>x1.39 | x2.45 | x1.52 | x1.64 | x1.48</td></tr>
<tr><td>summary: run all-flows, single hel.</td><td>x1.13 | x1.14 | x1.14 | x1.14 | x1.14</td><td>x0.899 | x1.25 | x1.00 | x1.04 | x1.06</td><td>x0.838 | x1.59 | x0.960 | x1.03 | x1.18</td></tr>
</table>

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol LC (block 2 of 3)

ID	base process	metric	n=4				n=5				n=6			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.84	0.00489	x2.08	15.5	8.16	0.0149	x2.15	15.7	8.78	0.140	x2.46	25.2
		run [μs/pt]	4.46	3.05	([x0.983] x1.08)	([x0.696] x0.864)	13.2	15.3	([x0.794] x0.852)	([x0.694] x0.750)	34.1	96.3	([x0.782] x0.821)	([x0.722] x0.748)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.83	0.00238	x2.42	16.2	8.15	0.0149	x2.02	17.4	8.40	0.149	x2.40	25.4
		run [μs/pt]	2.52	4.21	([x1.22] x1.65)	([x0.509] x0.626)	9.00	15.4	([x1.01] x1.04)	([x0.675] x0.735)	26.6	98.6	([x0.683] x0.721)	([x0.719] x0.741)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.42	0.00201	x1.63	15.1	9.89	0.0115	x1.55	14.9	11.2	0.0978	x1.59	16.2
		run [μs/pt]	1.24	0.964	([x1.12] x1.32)	([x0.577] x0.913)	3.85	3.70	([x1.02] x1.16)	([x0.669] x0.813)	11.1	16.7	([x0.842] x0.894)	([x0.684] x0.747)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.29	0.00197	x1.58	16.0	9.41	0.0114	x1.61	14.8	10.1	0.0971	x1.66	16.2
		run [μs/pt]	0.413	0.831	([x1.28] x1.83)	([x0.607] x1.02)	1.66	3.15	([x1.34] x1.51)	([x0.678] x0.849)	6.05	15.2	([x1.22] x1.25)	([x0.672] x0.738)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	0.00433	x1.58	15.1	11.5	0.0123	x1.60	15.5	13.4	0.102	x1.78	18.2
		run [μs/pt]	6.36	1.68	([x1.20] x1.31)	([x0.643] x0.910)	16.7	6.30	([x1.00] x1.05)	([x0.757] x0.875)	42.9	30.6	([x0.885] x0.932)	([x0.708] x0.756)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	30.8	0.00328	x0.918	17.4	64.5	0.0348	x4.00	20.9	186	0.296	x1.42	63.5
		run [μs/pt]	6.89	8.41	([x1.01] x1.12)	([x0.898] x0.980)	21.2	77.5	([x14.8] x6.24)	([x0.512] x0.544)	58.0	479	([x0.894] x0.921)	([x0.797] x0.804)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	17.6	0.00499	x1.01	14.8	23.6	0.0146	x1.06	17.0	33.2	0.137	x1.60	29.3
		run [μs/pt]	2.39	4.16	([x1.31] x1.54)	([x0.704] x0.813)	8.08	23.6	([x0.990] x1.13)	([x0.604] x0.649)	23.6	157	([x0.913] x0.992)	([x0.705] x0.717)
8	$gg \to gg + (n-2)g$	gen. [s]	15.8	0.00514	x1.53	18.2	18.3	0.0838	x2.79	43.9	27.8	3.09	x5.26	342
		run [μs/pt]	9.36	26.2	([x0.674] x0.730)	([x0.734] x0.777)	28.7	221	([x0.652] x0.685)	([x0.757] x0.781)	79.0	2.74e+03	([x0.640] x0.668)	([x1.20] x1.36)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.3	0.00206	x0.833	15.6	26.8	0.0112	x0.790	15.3	38.4	0.0927	x0.776	16.4
		run [μs/pt]	12.9	1.56	([x1.16] x1.22)	([x0.572] x0.898)	33.3	3.94	([x1.05] x1.08)	([x0.669] x0.878)	84.1	14.5	([x0.989] x1.05)	([x0.692] x0.799)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	57.0	0.00195	x0.268	15.2	61.7	0.0111	x0.249	14.5	65.9	0.0922	x0.255	15.0
		run [μs/pt]	2.06	1.29	([x1.46] x1.64)	([x0.572] x0.838)	4.66	1.92	([x1.22] x1.35)	([x0.602] x0.853)	10.7	3.90	([x1.05] x1.14)	([x0.650] x0.849)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	31.1	0.00201	x0.527	14.4	64.4	0.0104	x0.340	15.8	117	0.107	x0.331	16.6
		run [μs/pt]	2.99	1.63	([x1.48] x1.63)	([x0.688] x0.925)	9.01	4.83	([x1.33] x1.43)	([x0.717] x0.861)	26.0	36.5	([x1.21] x1.27)	([x0.375] x0.410)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	63.3	0.00432	x0.236	14.3	67.5	0.0110	x0.240	14.6	65.6	0.0913	x0.279	16.1
		run [μs/pt]	3.81	1.85	([x0.815] x0.912)	([x0.597] x0.780)	7.86	2.74	([x0.708] x0.788)	([x0.589] x0.772)	17.8	4.96	([x0.718] x0.774)	([x0.666] x0.812)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	12.4	0.00149	x1.29	15.2	21.5	0.00985	x0.896	16.2	31.6	0.0946	x2.00	20.6
		run [μs/pt]	0.457	2.07	([x1.39] x2.33)	([x0.790] x0.982)	1.14	9.33	([x1.26] x2.13)	([x0.807] x0.890)	3.05	50.9	([x1.18] x4.37)	([x0.885] x0.914)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]			NOT APPLICABLE				NOT APPLICABLE		STATIC N/A	STATIC N/A	24.5	18.0
		run [μs/pt]									STATIC N/A	STATIC N/A	5.61	23.1
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	11.9	0.00195	x1.31	15.7	18.7	0.0110	x1.03	17.2	26.4	0.101	x2.03	26.3
		run [μs/pt]	0.678	3.22	([x1.21] x2.02)	([x0.852] x1.01)	1.58	15.8	([x1.12] x1.95)	([x0.801] x0.868)	3.90	87.2	([x1.08] x1.81)	([x1.02] x1.05)

	n=4					n=5					n=6				
summary: generation	x0.236	x2.42	x1.30	x1.23	**$\boxed{\text{x0.810}}$**	x0.240	x4.00	x1.30	x1.45	**$\boxed{\text{x1.28}}$**	x0.255	x5.26	x1.63	x1.70	**$\boxed{\text{x1.22}}$**
summary: run single-flow, hel. sum	x0.730	x2.33	x1.43	x1.45	**$\boxed{\text{x1.20}}$**	x0.685	x6.24	x1.15	x1.60	**$\boxed{\text{x1.71}}$**	x0.668	x4.37	x0.962	x1.26	**$\boxed{\text{x0.942}}$**
summary: run all-flows, single hel.	x0.626	x1.02	x0.904	x0.881	**$\boxed{\text{x0.838}}$**	x0.544	x0.890	x0.831	x0.794	**$\boxed{\text{x0.735}}$**	x0.410	x1.36	x0.777	x0.818	**$\boxed{\text{x1.19}}$**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol LC (block 3 of 3)

ID	base process	metric	n=7			n=8			n=9		
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	9.93	1.45 x3.57	130	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	90.9	839 ([x0.749] x0.778)	([x1.04] x1.04)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.28	1.50 x3.56	132	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	77.2	941 ([x0.826] x0.653)	([x0.951] x0.950)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	11.1	1.01 x1.85	26.8	13.9	11.2 x2.97	131	20.0	138 x6.04	N/A
		run [μs/pt]	31.3	105 ([x0.745] x0.785)	([x0.692] x0.716)	82.7	888 ([x0.730] x0.755)	([x0.773] x0.780)	206	1.66e+04 ([x0.781] x0.809)	N/A
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	10.5	0.961 x1.77	25.5	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	19.0	96.2 ([x0.739] x0.793)	([x0.691] x0.716)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	20.4	1.10 x1.77	44.7	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	111	192 ([x0.838] x0.862)	([x0.754] x0.814)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	1.17e+03	5.67 x0.766	867	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	153	1.05e+04 ([x0.909] x0.946)	([x1.26] x1.25)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	60.7	1.60 x2.22	215	372	28.8 x1.12	N/A	3.78e+03	N/A N/A	N/A
		run [μs/pt]	65.1	1.51e+03 ([x0.919] x0.975)	([x1.36] x1.39)	171	2.61e+04 ([x0.987] x1.04)	N/A	429	N/A N/A	N/A
8	$gg \to gg + (n-2)g$	gen. [s]	55.4	216 x14.3	7.19e+03	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	210	6.68e+04 ([x0.634] x0.706)	([x1.63] x1.61)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	83.0	0.981 x0.596	29.0	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	205	67.0 ([x0.977] x1.00)	([x0.675] x0.721)	N/A	N/A N/A	N/A	N/A	N/A N/A	N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	75.9	0.953 x0.269	16.7	108	10.2 x0.233	21.7	164	120 x0.242	76.4
		run [μs/pt]	25.5	12.5 ([x0.981] x1.02)	([x0.682] x0.791)	61.4	55.2 ([x0.883] x0.918)	([x0.662] x0.712)	146	319 ([x0.884] x0.905)	([x0.799] x0.820)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	238	0.954 x0.419	25.3	794	N/A N/A	N/A	N/A	N/A N/A	N/A
		run [μs/pt]	70.9	109 ([x1.29] x1.34)	([x0.692] x0.717)	187	N/A N/A	N/A	N/A	N/A N/A	N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	78.7	0.900 x0.250	16.4	128	10.1 x0.204	24.8	181	119 x0.215	104
		run [μs/pt]	41.5	14.3 ([x0.578] x0.615)	([x0.762] x0.848)	152	65.6 ([x0.345] x0.356)	([x0.771] x0.896)	213	373 ([x0.556] x0.566)	([x0.803] x0.818)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	58.1	0.963 x3.41	71.9	143	14.7 x1.09	1.30e+03	360	149 x2.88	N/A
		run [μs/pt]	8.25	362 ([x1.34] x1.92)	([x1.49] x1.50)	21.3	4.15e+03 ([x1.96] x2.39)	([x3.63] x3.63)	54.4	5.80e+04 ([x3.55] x4.03)	N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	STATIC N/A	STATIC N/A 116	39.5	STATIC N/A	STATIC N/A 684	336	STATIC N/A	STATIC N/A BLOCKED DEP.	BLOCKED DEP.
		run [μs/pt]	STATIC N/A	STATIC N/A 11.0	189	STATIC N/A	STATIC N/A 104	2.64e+03	STATIC N/A	STATIC N/A BLOCKED DEP.	BLOCKED DEP.
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	43.7	1.02 x4.38	138	95.7	15.5 x1.70	3.49e+03	231	165 x4.46	BLOCKED DEP.
		run [μs/pt]	9.93	624 ([x2.69] x1.81)	([x2.26] x2.29)	27.2	1.29e+04 ([x1.50] x1.93)	([x4.76] x4.02)	77.1	1.49e+05 ([x2.38] x2.75)	BLOCKED DEP.

	n=7					n=8					n=9				
summary: generation	x0.250	x14.3	x1.81	x2.80	$\boxed{\text{x1.32}}$	x0.204	x2.97	x1.10	x1.22	$\boxed{\text{x0.961}}$	x0.215	x6.04	x2.88	x2.77	$\boxed{\text{x2.37}}$
summary: run single-flow, hel. sum	x0.615	x1.92	x0.904	x1.02	$\boxed{\text{x0.893}}$	x0.356	x2.39	x0.980	x1.23	$\boxed{\text{x0.882}}$	x0.566	x4.03	x0.905	x1.81	$\boxed{\text{x1.22}}$
summary: run all-flows, single hel.	x0.716	x2.29	x0.899	x1.10	$\boxed{\text{x1.55}}$	x0.712	x4.02	x0.896	x2.01	$\boxed{\text{x3.75}}$	x0.818	x0.820	x0.819	x0.819	$\boxed{\text{x0.819}}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.8 UFO-SM recurrence versus AmpliCol NLC

UFO-SM recurrence versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.10 x2.01	7.24 x2.12	7.21 x2.08
		run [μs/pt]	0.228 ([x0.798] x1.26)	0.668 ([x0.896] x1.17)	3.57 ([x0.719] x0.954)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.01 x2.20	7.77 x1.88	7.40 x2.07
		run [μs/pt]	0.138 ([x0.915] x1.46)	0.376 ([x0.962] x1.30)	1.93 ([x0.732] x1.39)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 x1.34	9.31 x1.56
		run [μs/pt]		0.466 ([x0.574] x0.790)	1.11 ([x0.536] x0.685)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	8.95 x1.61	9.13 x1.59
		run [μs/pt]		0.246 ([x0.356] x0.648)	0.505 ([x0.389] x0.612)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.95 x1.46	10.1 x1.48
		run [μs/pt]		0.936 ([x0.909] x1.19)	2.85 ([x0.949] x1.12)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.99 x1.66	17.0 x0.995
		run [μs/pt]		1.71 ([x0.650] x0.992)	16.5 ([x0.597] x0.902)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.35 x1.60	14.0 x1.10
		run [μs/pt]		1.17 ([x0.349] x0.614)	6.25 ([x0.405] x0.635)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.0 x1.34	13.2 x1.36
		run [μs/pt]		7.93 ([x0.560] x0.759)	120 ([x0.355] x0.470)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.4 x0.807
		run [μs/pt]			5.70 ([x1.01] x1.15)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1	n=2	n=3
summary: generation	x2.01 | x2.20 | x2.11 | x2.11 | **x2.11**	x1.34 | x2.12 | x1.61 | x1.63 | **x1.59**	x0.807 | x2.08 | x1.48 | x1.45 | **x1.31**
summary: wall	x1.26 | x1.46 | x1.36 | x1.36 | **x1.33**	x0.614 | x1.30 | x0.891 | x0.933 | **x0.840**	x0.470 | x1.39 | x0.902 | x0.881 | **x0.582**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol NLC (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.19	x2.45	8.34	x7.46	NOT RUN	
		run [μs/pt]	31.4	([x0.560] x0.800)	366	([x0.581] x0.833)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.70	x3.15	7.64	x5.65	NOT RUN	
		run [μs/pt]	29.9	([x0.321] x0.453)	250	([x0.526] x0.601)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.43	x1.60	10.3	x1.88	NOT RUN	
		run [μs/pt]	5.53	([x0.420] x0.540)	44.7	([x0.340] x0.455)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.44	x1.56	9.51	x1.71	NOT RUN	
		run [μs/pt]	2.50	([x0.338] x0.454)	22.3	([x0.310] x0.379)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.6	x1.61	11.7	x2.87	NOT RUN	
		run [μs/pt]	15.8	([x0.765] x0.915)	117	([x0.771] x0.969)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	33.1	x1.15	63.4	x12.1	NOT RUN	
		run [μs/pt]	230	([x0.516] x0.767)	3.46e+03	([x2.38] x2.90)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	18.7	x1.09	28.1	x2.41	NOT RUN	
		run [μs/pt]	57.7	([x0.365] x0.538)	695	([x0.550] x0.820)		
8	$gg \to gg + (n-2)g$	gen. [s]	17.0	x1.57	22.1	x2.70	NOT RUN	
		run [μs/pt]	2.01e+03	([x0.352] x0.543)	3.66e+04	([x0.435] x0.905)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.2	x0.809	26.7	x1.00	NOT RUN	
		run [μs/pt]	15.0	([x0.992] x1.08)	74.7	([x0.857] x0.958)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	58.9	x0.259	63.6	x0.241	NOT RUN	
		run [μs/pt]	3.91	([x0.766] x0.861)	8.84	([x0.621] x0.691)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.7	x0.476	69.2	x0.290	NOT RUN	
		run [μs/pt]	11.8	([x0.602] x0.692)	65.1	([x0.586] x0.672)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6	x0.228	72.0	x0.225	NOT RUN	
		run [μs/pt]	7.35	([x0.406] x0.462)	19.4	([x0.313] x0.344)		
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	0.00152	x1.05e+04	0.0101	x2.10e+03	0.101	x901
		run [μs/pt]	144	([x0.0174] x0.0254)	1.28e+03	([x0.0157] x0.0242)	1.49e+04	([x0.0239] x0.0314)
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		STATIC N/A	40.8
		run [μs/pt]					STATIC N/A	79.1
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	0.00198	x8.16e+03	0.0112	x2.37e+03	0.108	x1.37e+03
		run [μs/pt]	219	([x0.0193] x0.0267)	2.02e+03	([x0.0189] x0.0264)	2.40e+04	([x0.0411] x0.0477)

	n=4					n=5					n=6				
summary: generation	x0.228	x1.05e+04	x1.57	x1.33e+03	x0.927	x0.225	x2.37e+03	x2.55	x322	x3.05	x901	x1.37e+03	x1.14e+03	x1.14e+03	x1.14e+03
summary: wall	x0.0254	x1.08	x0.542	x0.583	x0.502	x0.0242	x2.90	x0.681	x0.756	x0.989	x0.0314	x0.0477	x0.0395	x0.0395	x0.0414

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.9 UFO-SM recurrence versus AmpliCol full-colour

UFO-SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	7.17 x1.99	7.21 x2.13	7.26 x2.20
		run [μs/pt]	0.270 ([x0.671] x1.04)	0.744 ([x0.837] x1.09)	3.97 ([x0.646] x0.866)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.45 x2.11	7.59 x2.01	7.90 x2.26
		run [μs/pt]	0.159 ([x0.794] x1.25)	0.428 ([x0.805] x1.11)	2.14 ([x0.615] x0.985)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	9.28 x1.61	9.55 x1.52
		run [μs/pt]		0.522 ([x0.519] x0.715)	1.20 ([x0.481] x0.618)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	163 x0.0899	9.24 x1.56
		run [μs/pt]		0.365 ([x0.240] x0.443)	0.559 ([x0.348] x0.551)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	9.91 x1.52	10.2 x1.52
		run [μs/pt]		1.05 ([x0.804] x1.05)	3.08 ([x0.906] x1.07)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.91 x1.67	16.5 x1.06
		run [μs/pt]		1.98 ([x0.581] x0.882)	19.5 ([x0.492] x0.846)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.57 x1.55	13.7 x1.11
		run [μs/pt]		1.48 ([x0.282] x0.487)	7.52 ([x0.340] x0.529)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	11.8 x1.25	13.6 x1.27
		run [μs/pt]		9.31 ([x0.375] x0.547)	146 ([x0.288] x0.395)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	19.3 x0.800
		run [μs/pt]			6.11 ([x0.958] x1.09)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1	n=2	n=3
summary: generation	x1.99 | x2.11 | x2.05 | x2.05 | **x2.05**	x0.0899 | x2.13 | x1.58 | x1.48 | **x0.526**	x0.800 | x2.26 | x1.52 | x1.48 | **x1.34**
summary: wall	x1.04 | x1.25 | x1.14 | x1.14 | **x1.12**	x0.443 | x1.11 | x0.799 | x0.791 | **x0.660**	x0.395 | x1.09 | x0.846 | x0.773 | **x0.499**

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric	n=4	n=5	n=6
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.15 x2.46	8.47 x7.28	NOT RUN
		run [μs/pt]	36.1 ([x0.488] x0.775)	464 ([x0.458] x0.655)	
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	7.75 x2.75	8.32 x5.08	NOT RUN
		run [μs/pt]	20.6 ([x0.564] x0.716)	283 ([x0.363] x0.525)	
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	9.59 x1.61	9.87 x1.86	NOT RUN
		run [μs/pt]	6.11 ([x0.406] x0.508)	51.6 ([x0.295] x0.434)	
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	9.27 x1.59	9.73 x1.86	NOT RUN
		run [μs/pt]	2.80 ([x0.261] x0.366)	25.4 ([x0.278] x0.362)	
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5 x1.70	11.5 x2.95	NOT RUN
		run [μs/pt]	16.7 ([x0.723] x0.865)	131 ([x0.689] x0.930)	
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	31.2 x1.22	63.2 x16.6	NOT RUN
		run [μs/pt]	283 ([x0.431] x0.658)	6.16e+03 ([x1.23] x1.71)	
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	19.3 x1.08	28.0 x2.45	NOT RUN
		run [μs/pt]	74.3 ([x0.287] x0.412)	1.09e+03 ([x0.353] x0.579)	
8	$gg \to gg + (n-2)g$	gen. [s]	16.3 x1.70	22.2 x2.99	NOT RUN
		run [μs/pt]	3.06e+03 ([x0.233] x0.424)	9.67e+04 ([x0.133] x0.350)	
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	21.5 x0.831	26.8 x0.999	NOT RUN
		run [μs/pt]	15.8 ([x0.930] x1.02)	79.5 ([x0.813] x0.905)	
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	59.5 x0.264	63.8 x0.242	NOT RUN
		run [μs/pt]	4.13 ([x0.702] x0.792)	9.25 ([x0.597] x0.661)	
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	32.5 x0.506	69.4 x0.302	NOT RUN
		run [μs/pt]	12.8 ([x0.569] x0.649)	68.6 ([x0.558] x0.627)	
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	65.6 x0.233	71.9 x0.216	NOT RUN
		run [μs/pt]	7.59 ([x0.405] x0.456)	20.0 ([x0.305] x0.337)	
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	0.00151 x1.02e+04	0.0101 x2.14e+03	0.104 x777
		run [μs/pt]	150 ([x0.0166] x0.0243)	1.31e+03 ([x0.0155] x0.0235)	1.64e+04 ([x0.0223] x0.0294)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	STATIC N/A 40.4
		run [μs/pt]			STATIC N/A 79.5
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	0.00198 x8.59e+03	0.0112 x2.37e+03	0.110 x1.36e+03
		run [μs/pt]	225 ([x0.0188] x0.0262)	2.07e+03 ([x0.0183] x0.0258)	2.59e+04 ([x0.0381] x0.0448)

	n=4	n=5	n=6
summary: generation	x0.233 | x1.02e+04 | x1.60 | x1.34e+03 | $\boxed{\text{x0.940}}$	x0.216 | x2.37e+03 | x2.70 | x325 | $\boxed{\text{x3.78}}$	x777 | x1.36e+03 | x1.07e+03 | x1.07e+03 | $\boxed{\text{x1.08e+03}}$
summary: wall	x0.0243 | x1.02 | x0.579 | x0.550 | $\boxed{\text{x0.413}}$	x0.0235 | x1.71 | x0.552 | x0.580 | $\boxed{\text{x0.422}}$	x0.0294 | x0.0448 | x0.0371 | x0.0371 | $\boxed{\text{x0.0388}}$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.10 Built-in SM compiled JIT O3 versus recurrence JIT O2 LC

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 1 of 3)

ID	base process	metric	n=1				n=2				n=3			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.31	8.12	x0.521	x0.471	8.91	8.21	x0.519	x0.488	8.76	8.20	x0.779	x0.591
		run [μs/pt]	0.245	0.142	([x1.42] x1.42)	([x1.26] x1.26)	0.663	0.301	([x1.07] x1.07)	([x1.07] x1.07)	1.62	0.706	([x0.757] x0.756)	([x0.753] x0.753)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.05	8.05	x0.510	x0.443	8.45	9.16	x0.524	x0.409	9.79	9.32	x0.679	x0.474
		run [μs/pt]	0.188	0.137	([x1.43] x1.43)	([x1.27] x1.27)	0.489	0.299	([x1.10] x1.10)	([x2.45] x2.44)	1.18	0.701	([x0.970] x0.970)	([x0.751] x0.751)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE				8.17	8.27	x0.622	x0.563	8.83	8.25	x0.666	x0.621
		run [μs/pt]					0.347	0.243	([x1.11] x1.11)	([x1.11] x1.11)	0.685	0.427	([x1.01] x1.01)	([x0.892] x0.891)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE				8.54	8.47	x0.495	x0.499	8.30	8.75	x0.550	x0.502
		run [μs/pt]					0.184	0.213	([x1.18] x1.18)	([x1.20] x1.20)	0.333	0.363	([x1.25] x1.25)	([x1.05] x1.05)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE				8.28	8.40	x0.579	x0.477	9.13	8.74	x0.618	x0.497
		run [μs/pt]					0.939	0.305	([x0.914] x0.914)	([x1.07] x1.06)	2.50	0.653	([x0.816] x0.816)	([x0.755] x0.754)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.85	9.16	x0.552	x0.427	10.9	8.31	x0.953	x0.671
		run [μs/pt]					0.843	0.437	([x0.866] x0.866)	([x0.853] x0.852)	3.23	1.45	([x0.582] x0.581)	([x0.751] x0.751)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.22	8.96	x0.588	x0.499	9.27	8.27	x0.815	x0.687
		run [μs/pt]					0.402	0.256	([x1.23] x1.23)	([x1.14] x1.14)	1.10	0.742	([x0.907] x0.907)	([x0.811] x0.811)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE				8.93	8.22	x0.602	x0.513	10.7	9.13	x1.05	x0.738
		run [μs/pt]					0.725	0.864	([x0.888] x0.888)	([x0.505] x0.504)	3.22	5.00	([x0.633] x0.633)	([x0.378] x0.377)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				9.75	8.74	x0.743	x0.532
		run [μs/pt]									5.28	0.761	([x0.815] x0.815)	([x0.787] x0.787)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												

	n=1					n=2					n=3				
summary: generation	x0.510	x0.521	x0.516	x0.516	**x0.516**	x0.495	x0.622	x0.565	x0.560	**x0.560**	x0.550	x1.05	x0.743	x0.762	**x0.773**
summary: run single-flow, hel. sum	x1.42	x1.43	x1.43	x1.43	**x1.43**	x0.866	x1.23	x1.08	x1.05	**x0.997**	x0.581	x1.25	x0.816	x0.860	**x0.769**
summary: run all-flows, single hel.	x1.26	x1.27	x1.26	x1.26	**x1.26**	x0.504	x2.44	x1.09	x1.17	**x1.03**	x0.377	x1.05	x0.754	x0.770	**x0.600**

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 2 of 3)

ID	base process	metric	n=4				n=5				n=6			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	9.62	8.33	x1.29	x1.12	10.5	9.44	x3.58	x3.25	20.5	17.9	x11.2	x12.4
		run [μs/pt]	4.97	2.45	([x0.651] x0.650)	([x0.559] x0.558)	10.8	10.8	([x0.918] x0.917)	([x0.648] x0.647)	50.4	66.0	([x0.752] x0.752)	([x1.06] x1.06)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	10.2	9.83	x1.13	x0.821	10.3	10.7	x2.74	x1.86	13.4	19.2	x11.4	x7.41
		run [μs/pt]	4.06	2.53	([x0.537] x0.537)	([x0.610] x0.609)	9.53	11.3	([x0.740] x0.740)	([x0.613] x0.612)	18.3	69.6	([x1.37] x1.37)	([x1.00] x1.00)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.88	8.30	x1.03	x0.809	9.10	8.37	x1.95	x1.62	10.5	10.3	x5.10	x4.33
		run [μs/pt]	1.52	0.880	([x0.774] x0.774)	([x0.719] x0.719)	4.62	2.87	([x0.639] x0.639)	([x0.547] x0.546)	9.71	12.4	([x0.925] x0.924)	([x0.618] x0.617)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.86	8.33	x0.743	x0.537	8.63	9.08	x1.08	x0.831	9.42	9.55	x2.51	x2.05
		run [μs/pt]	0.727	0.797	([x1.03] x1.03)	([x0.759] x0.759)	2.34	3.43	([x0.719] x0.719)	([x0.432] x0.432)	7.92	11.3	([x0.658] x0.658)	([x0.643] x0.642)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	9.43	8.22	x1.04	x0.762	11.2	8.79	x1.97	x1.53	15.7	12.0	x5.23	x5.03
		run [μs/pt]	6.51	1.39	([x0.680] x0.679)	([x0.650] x0.649)	15.5	5.12	([x0.843] x0.843)	([x0.484] x0.484)	36.3	21.3	([x1.22] x1.21)	([x0.559] x0.558)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	20.8	9.07	x1.75	x2.00	74.3	13.7	x12.2	x6.54	241	59.9	x1.65	x15.3
		run [μs/pt]	9.54	8.00	([x0.647] x0.647)	([x0.905] x0.905)	18.8	66.7	([x1.33] x1.33)	([x1.00] x1.00)	62.2	487	([x1.37] x1.37)	([x1.40] x1.40)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	10.7	8.93	x1.97	x1.22	17.3	9.93	x3.61	x3.77	43.5	22.7	x1.73	x10.4
		run [μs/pt]	3.86	3.21	([x0.583] x0.583)	([x0.582] x0.582)	8.67	14.4	([x0.838] x0.838)	([x0.716] x0.715)	22.1	107	([x1.20] x1.20)	([x0.915] x0.914)
8	$gg \to gg + (n-2)g$	gen. [s]	18.1	11.4	x2.35	x2.12	52.4	37.3	x1.17	x5.93	134	319	x1.42	x9.01
		run [μs/pt]	7.11	20.4	([x1.14] x1.14)	([x0.789] x0.788)	19.6	172	([x1.97] x1.97)	([x1.17] x1.17)	52.6	2.99e+03	([x2.26] x2.25)	([x0.758] x0.758)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	10.6	8.36	x1.02	x0.704	13.9	8.58	x1.81	x1.03	21.4	10.8	x3.06	x2.89
		run [μs/pt]	13.0	1.39	([x0.916] x0.916)	([x0.681] x0.681)	29.7	3.33	([x1.24] x1.24)	([x0.535] x0.534)	70.7	10.9	([x1.33] x1.33)	([x0.457] x0.456)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	9.37	9.17	x0.852	x0.674	9.00	8.32	x1.13	x0.889	10.5	8.83	x2.52	x1.25
		run [μs/pt]	2.97	1.07	([x0.698] x0.697)	([x0.734] x0.734)	5.34	1.60	([x0.754] x0.754)	([x0.683] x0.682)	10.5	3.30	([x0.814] x0.813)	([x0.563] x0.563)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.30	8.77	x0.951	x0.693	14.9	8.95	x1.61	x1.13	30.3	9.97	x3.22	x3.37
		run [μs/pt]	4.64	1.43	([x0.639] x0.639)	([x0.689] x0.689)	11.9	3.99	([x0.757] x0.756)	([x0.637] x0.636)	30.9	14.4	([x1.15] x1.15)	([x0.705] x0.704)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	9.17	8.90	x1.42	x1.29	9.59	8.39	x1.77	x1.67	10.1	8.82	x4.03	x2.66
		run [μs/pt]	3.11	1.44	([x0.711] x0.710)	([x0.617] x0.617)	5.66	1.98	([x0.753] x0.752)	([x0.639] x0.638)	11.0	3.85	([x0.795] x0.794)	([x0.545] x0.544)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	9.26	8.64	x1.46	x1.21	12.5	9.72	x3.27	x2.79	43.6	14.0	x5.98	x8.26
		run [μs/pt]	1.06	1.92	([x0.804] x0.803)	([x0.546] x0.546)	2.38	7.90	([x0.650] x0.650)	([x0.488] x0.487)	5.83	46.2	([x0.508] x0.508)	([x0.464] x0.464)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				17.4	11.8	x7.60	x8.07
		run [μs/pt]									7.05	22.0	([x0.711] x0.710)	([x0.492] x0.492)
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	9.42	9.18	x2.67	x1.70	12.2	10.4	x6.03	x4.11	40.8	17.9	x7.31	x11.8
		run [μs/pt]	1.77	3.20	([x0.549] x0.549)	([x0.486] x0.485)	3.04	13.5	([x0.563] x0.562)	([x0.477] x0.477)	6.92	92.0	([x0.524] x0.523)	([x0.435] x0.435)

	n=4					n=5					n=6				
summary: generation	x0.743	x2.67	x1.21	x1.40	$\boxed{\text{x1.49}}$	x1.08	x12.2	x1.96	x3.14	$\boxed{\text{x5.03}}$	x1.42	x11.4	x4.03	x4.93	$\boxed{\text{x3.21}}$
summary: run single-flow, hel. sum	x0.537	x1.14	x0.688	x0.740	$\boxed{\text{x0.760}}$	x0.562	x1.97	x0.755	x0.908	$\boxed{\text{x1.10}}$	x0.508	x2.25	x0.924	x1.04	$\boxed{\text{x1.27}}$
summary: run all-flows, single hel.	x0.485	x0.905	x0.665	x0.666	$\boxed{\text{x0.727}}$	x0.432	x1.17	x0.624	x0.647	$\boxed{\text{x0.986}}$	x0.435	x1.40	x0.617	x0.707	$\boxed{\text{x0.835}}$

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 3 of 3)

ID	base process	metric	n=7				n=8				n=9			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	27.1	113	x1.68	x19.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	68.1	804	([x1.70] x1.70)	([x0.971] x0.969)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	24.7	123	x1.58	x12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	58.7	890	([x1.51] x1.51)	([x0.858] x0.857)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	13.8	19.4	x22.3	x14.8	25.5	130	N/A	N/A	104	3.79e+03	N/A	N/A
		run [μs/pt]	23.5	74.1	([x1.33] x1.33)	([x0.989] x0.988)	65.6	692	N/A	N/A	155	2.65e+04	N/A	N/A
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	11.9	19.0	x10.4	x5.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	14.9	69.5	([x1.37] x1.37)	([x0.974] x0.972)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	26.8	36.3	x21.1	x14.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	86.9	141	([x1.45] x1.45)	([x0.784] x0.783)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	847	822	x1.62	ERROR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	144	1.25e+04	([x1.60] x1.60)	ERROR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	126	178	x1.58	x12.1	383	4.69e+03	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	64.7	1.85e+03	([x1.47] x1.47)	([x0.505] x0.505)	173	7.12e+04	N/A	N/A	N/A	N/A	N/A	N/A
8	$gg \to gg + (n-2)g$	gen. [s]	780	6.59e+03	x0.932	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	145	1.24e+05	([x2.32] x2.29)	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	38.3	21.5	x7.05	x8.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	168	43.7	([x1.44] x1.44)	([x0.549] x0.548)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	13.2	9.59	x3.17	x2.73	18.2	15.1	N/A	N/A	30.5	64.3	N/A	N/A
		run [μs/pt]	23.7	9.89	([x0.856] x0.855)	([x0.475] x0.474)	49.8	38.0	N/A	N/A	118	245	N/A	N/A
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	94.6	18.0	x3.88	x8.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	84.1	75.4	([x1.29] x1.29)	([x0.941] x0.940)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.4	10.6	x6.33	x5.47	17.5	18.2	N/A	N/A	31.9	94.6	N/A	N/A
		run [μs/pt]	23.1	11.5	([x0.933] x0.932)	([x0.463] x0.463)	47.7	41.0	N/A	N/A	113	273	N/A	N/A
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	180	59.8	x2.13	x10.8	126	1.19e+03	N/A	N/A	961	N/A	N/A	N/A
		run [μs/pt]	22.2	475	([x1.31] x1.30)	([x0.365] x0.364)	50.6	1.41e+04	N/A	N/A	220	N/A	N/A	N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	100	29.8	x7.30	x15.6	592	287	N/A	N/A	ERROR	BLOCKED DEP.	N/A	N/A
		run [μs/pt]	10.6	175	([x1.46] x1.46)	([x0.360] x0.360)	99.0	2.49e+03	N/A	N/A	ERROR	BLOCKED DEP.	N/A	N/A
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	158	119	x2.28	x9.45	143	3.45e+03	N/A	N/A	951	ERROR	N/A	N/A
		run [μs/pt]	17.6	1.41e+03	([x2.13] x2.12)	([x0.237] x0.236)	75.5	5.11e+04	N/A	N/A	219	ERROR	N/A	N/A

	n=7					n=8	n=9
summary: generation	x0.932	x22.3	x3.17	x6.23	$\boxed{\text{x2.29}}$	N/A	N/A
summary: run single-flow, hel. sum	x0.855	x2.29	x1.45	x1.47	$\boxed{\text{x1.59}}$	N/A	N/A
summary: run all-flows, single hel.	x0.236	x0.988	x0.548	x0.651	$\boxed{\text{x0.564}}$	N/A	N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.11 Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.24 x0.501	8.28 x0.536	9.18 x0.579
		run [μs/pt]	0.267 ([x1.19] x1.19)	0.732 ([x0.864] x0.864)	3.23 ([x0.626] x0.626)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.39 x0.453	8.35 x0.499	9.01 x0.602
		run [μs/pt]	0.193 ([x1.59] x1.59)	0.491 ([x1.12] x1.12)	1.82 ([x0.865] x0.864)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	8.20 x0.613	8.28 x0.668
		run [μs/pt]		0.341 ([x1.05] x1.05)	0.731 ([x0.905] x0.905)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	8.47 x0.465	8.77 x0.528
		run [μs/pt]		0.170 ([x1.75] x1.75)	0.300 ([x1.77] x1.77)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	8.87 x0.510	9.13 x0.552
		run [μs/pt]		1.01 ([x0.775] x0.775)	2.77 ([x0.650] x0.650)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.64 x0.457	10.4 x0.754
		run [μs/pt]		1.69 ([x0.601] x0.601)	14.3 ([x0.618] x0.618)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.88 x0.526	8.82 x0.816
		run [μs/pt]		0.711 ([x2.21] x2.21)	3.90 ([x1.70] x1.70)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	9.16 x0.474	11.6 x0.632
		run [μs/pt]		5.01 ([x0.527] x0.527)	56.0 ([x0.916] x0.915)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	9.27 x0.638
		run [μs/pt]			6.17 ([x0.624] x0.624)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1	n=2	n=3
summary: generation	x0.453 | x0.501 | x0.477 | x0.477 | $\boxed{\text{x0.477}}$	x0.457 | x0.613 | x0.504 | x0.510 | $\boxed{\text{x0.508}}$	x0.528 | x0.816 | x0.632 | x0.641 | $\boxed{\text{x0.642}}$
summary: wall	x1.19 | x1.59 | x1.39 | x1.39 | $\boxed{\text{x1.36}}$	x0.527 | x2.21 | x0.955 | x1.11 | $\boxed{\text{x0.773}}$	x0.618 | x1.77 | x0.864 | x0.963 | $\boxed{\text{x0.865}}$

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 2 of 2)

ID	base process	metric	n=4	n=5	n=6
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	12.3 x0.941	48.1 x1.61	NOT RUN
		run [μs/pt]	23.7 ([x0.839] x0.839)	285 ([x1.01] x1.01)	
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	14.4 x0.723	36.3 x1.53	NOT RUN
		run [μs/pt]	14.6 ([x0.936] x0.935)	152 ([x1.34] x1.33)	
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.60 x0.836	12.2 x1.10	NOT RUN
		run [μs/pt]	2.84 ([x0.640] x0.639)	19.3 ([x0.837] x0.837)	
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.99 x0.523	10.0 x0.763	NOT RUN
		run [μs/pt]	0.996 ([x1.07] x1.07)	8.41 ([x0.862] x0.861)	
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.7 x0.754	25.1 x1.22	NOT RUN
		run [μs/pt]	12.7 ([x0.703] x0.703)	100 ([x0.943] x0.942)	
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	27.9 x1.60	266 x3.99	NOT RUN
		run [μs/pt]	165 ([x0.879] x0.878)	9.19e+03 ([x0.282] x0.282)	
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	13.0 x1.97	53.1 x4.42	NOT RUN
		run [μs/pt]	30.1 ([x2.16] x2.16)	548 ([x1.71] x1.71)	
8	$gg \to gg + (n-2)g$	gen. [s]	20.0 x0.923	61.9 x1.37	NOT RUN
		run [μs/pt]	1.10e+03 ([x1.29] x1.29)	2.15e+04 ([x1.31] x1.31)	
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	11.2 x0.765	18.8 x1.17	NOT RUN
		run [μs/pt]	13.4 ([x0.829] x0.829)	59.6 ([x1.14] x1.14)	
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	9.16 x0.778	9.44 x0.905	NOT RUN
		run [μs/pt]	2.96 ([x0.655] x0.655)	5.31 ([x0.713] x0.713)	
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.34 x1.05	13.2 x2.44	NOT RUN
		run [μs/pt]	7.63 ([x1.74] x1.74)	40.1 ([x2.32] x2.32)	
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	8.60 x1.49	9.00 x1.75	NOT RUN
		run [μs/pt]	3.14 ([x0.665] x0.665)	5.71 ([x0.700] x0.699)	
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	9.59 x1.08	14.4 x2.36	66.5 x4.24
		run [μs/pt]	3.59 ([x1.27] x1.26)	39.0 ([x0.926] x0.925)	431 ([x1.09] x1.09)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	31.0 x5.84
		run [μs/pt]			76.6 ([x1.34] x1.34)
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	9.47 x1.56	20.6 x2.80	123 x3.93
		run [μs/pt]	6.09 ([x1.10] x1.10)	52.5 ([x1.06] x1.06)	1.05e+03 ([x0.790] x0.790)

	n=4	n=5	n=6
summary: generation	x0.523 | x1.97 | x0.932 | x1.07 | **x1.12**	x0.763 | x4.42 | x1.57 | x1.96 | **x2.90**	x3.93 | x5.84 | x4.24 | x4.67 | **x4.29**
summary: wall	x0.639 | x2.16 | x0.907 | x1.05 | **x1.23**	x0.282 | x2.32 | x0.978 | x1.08 | **x1.02**	x0.790 | x1.34 | x1.09 | x1.07 | **x0.899**

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.12 Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.12 x0.510	8.20 x0.524	8.99 x0.600
		run [μs/pt]	0.265 ([x1.15] x1.15)	0.715 ([x0.886] x0.886)	3.36 ([x0.596] x0.596)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.18 x0.462	8.21 x0.531	9.46 x0.682
		run [μs/pt]	0.196 ([x1.56] x1.56)	0.475 ([x1.16] x1.16)	1.83 ([x1.03] x1.03)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	8.92 x0.532	9.11 x0.665
		run [μs/pt]		0.339 ([x1.05] x1.05)	0.727 ([x0.897] x0.897)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	9.14 x0.433	9.16 x0.497
		run [μs/pt]		0.163 ([x1.90] x1.90)	0.372 ([x1.32] x1.32)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	8.31 x0.523	9.17 x0.556
		run [μs/pt]		1.09 ([x0.696] x0.695)	2.67 ([x0.678] x0.678)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.06 x0.508	10.6 x0.737
		run [μs/pt]		1.80 ([x0.592] x0.592)	17.2 ([x0.533] x0.532)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.52 x0.580	9.55 x0.784
		run [μs/pt]		0.718 ([x2.19] x2.19)	3.90 ([x1.71] x1.71)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	9.20 x0.474	11.4 x0.654
		run [μs/pt]		5.19 ([x0.493] x0.493)	57.8 ([x0.896] x0.896)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	9.62 x0.638
		run [μs/pt]			5.73 ([x0.684] x0.683)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1	n=2	n=3
summary: generation	x0.462 | x0.510 | x0.486 | x0.486 | $\boxed{\text{x0.486}}$	x0.433 | x0.580 | x0.524 | x0.513 | $\boxed{\text{x0.512}}$	x0.497 | x0.784 | x0.654 | x0.646 | $\boxed{\text{x0.648}}$
summary: wall	x1.15 | x1.56 | x1.36 | x1.36 | $\boxed{\text{x1.33}}$	x0.493 | x2.19 | x0.967 | x1.12 | $\boxed{\text{x0.744}}$	x0.532 | x1.71 | x0.896 | x0.927 | $\boxed{\text{x0.837}}$

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	12.6	x0.960	48.4	x1.61	NOT RUN	
		run [μs/pt]	26.7	([x0.751] x0.751)	284	([x1.07] x1.07)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	10.8	x0.992	33.2	x1.71	NOT RUN	
		run [μs/pt]	14.8	([x0.758] x0.757)	151	([x1.36] x1.36)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.81	x0.775	11.8	x1.14	NOT RUN	
		run [μs/pt]	2.73	([x0.667] x0.667)	21.2	([x0.769] x0.769)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.94	x0.545	9.47	x0.815	NOT RUN	
		run [μs/pt]	1.00	([x1.16] x1.16)	8.91	([x0.820] x0.819)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5	x0.815	24.4	x1.27	NOT RUN	
		run [μs/pt]	13.1	([x0.696] x0.696)	108	([x0.887] x0.886)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	29.8	x1.51	268	x5.08	NOT RUN	
		run [μs/pt]	187	([x0.835] x0.834)	1.00e+04	([x0.367] x0.367)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	13.3	x1.95	53.1	x4.43	NOT RUN	
		run [μs/pt]	29.1	([x2.21] x2.21)	597	([x1.69] x1.69)		
8	$gg \to gg + (n-2)g$	gen. [s]	20.2	x0.998	69.9	x2.40	NOT RUN	
		run [μs/pt]	1.27e+03	([x1.11] x1.11)	3.60e+04	([x0.825] x0.825)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	11.1	x0.774	18.7	x1.19	NOT RUN	
		run [μs/pt]	13.2	([x0.845] x0.845)	59.0	([x1.13] x1.13)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	8.71	x0.826	9.40	x0.934	NOT RUN	
		run [μs/pt]	2.90	([x0.667] x0.667)	6.18	([x0.618] x0.618)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.16	x1.06	13.6	x2.37	NOT RUN	
		run [μs/pt]	8.00	([x1.65] x1.65)	39.4	([x2.31] x2.31)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	8.87	x1.41	9.65	x1.70	NOT RUN	
		run [μs/pt]	3.08	([x0.708] x0.708)	5.59	([x0.725] x0.724)		
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	9.47	x1.11	14.4	x2.38	67.9	x4.01
		run [μs/pt]	3.52	([x1.32] x1.31)	30.5	([x1.14] x1.14)	434	([x1.16] x1.16)
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		30.8	x5.92
		run [μs/pt]					77.2	([x1.34] x1.33)
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	10.7	x1.53	20.7	x2.79	130	x3.89
		run [μs/pt]	7.48	([x0.893] x0.893)	52.1	([x1.05] x1.05)	1.17e+03	([x0.790] x0.790)

	n=4					n=5					n=6				
summary: generation	x0.545	x1.95	x0.995	x1.09	$\boxed{\text{x1.15}}$	x0.815	x5.08	x1.71	x2.13	$\boxed{\text{x3.51}}$	x3.89	x5.92	x4.01	x4.61	$\boxed{\text{x4.20}}$
summary: wall	x0.667	x2.21	x0.839	x1.02	$\boxed{\text{x1.09}}$	x0.367	x2.31	x0.970	x1.05	$\boxed{\text{x0.744}}$	x0.790	x1.33	x1.16	x1.09	$\boxed{\text{x0.909}}$

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.13 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 1 of 3)

ID	base process	metric	n=1				n=2				n=3			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.31	8.12	x0.512	x0.484	8.91	8.21	x0.452	x0.485	8.76	8.20	x0.498	x0.530
		run [μs/pt]	0.245	0.142	([x2.23] x2.23)	([x2.73] x2.73)	0.663	0.301	([x1.85] x1.85)	([x2.08] x2.08)	1.62	0.706	([x1.44] x1.44)	([x1.57] x1.57)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.05	8.05	x0.492	x0.492	8.45	9.16	x0.495	x0.449	9.79	9.32	x0.583	x0.468
		run [μs/pt]	0.188	0.137	([x2.59] x2.59)	([x2.79] x2.78)	0.489	0.299	([x2.23] x2.23)	([x2.05] x2.05)	1.18	0.701	([x1.49] x1.49)	([x1.48] x1.48)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE				8.17	8.27	x0.503	x0.506	8.83	8.25	x0.470	x0.541
		run [μs/pt]					0.347	0.243	([x1.77] x1.77)	([x1.92] x1.92)	0.685	0.427	([x1.69] x1.69)	([x1.59] x1.59)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE				8.54	8.47	x0.476	x0.471	8.30	8.75	x0.488	x0.474
		run [μs/pt]					0.184	0.213	([x2.37] x2.37)	([x2.03] x2.03)	0.333	0.363	([x2.05] x2.04)	([x1.77] x1.77)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE				8.28	8.40	x0.486	x0.479	9.13	8.74	x0.457	x0.477
		run [μs/pt]					0.939	0.305	([x1.76] x1.76)	([x2.35] x2.35)	2.50	0.653	([x1.68] x1.68)	([x1.57] x1.57)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.85	9.16	x0.454	x0.464	10.9	8.31	x0.439	x0.608
		run [μs/pt]					0.843	0.437	([x1.45] x1.45)	([x1.91] x1.91)	3.23	1.45	([x1.14] x1.14)	([x1.70] x1.70)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE				8.22	8.96	x0.524	x0.463	9.27	8.27	x0.529	x0.528
		run [μs/pt]					0.402	0.256	([x2.05] x2.05)	([x2.40] x2.40)	1.10	0.742	([x1.91] x1.91)	([x1.70] x1.70)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE				8.93	8.22	x0.453	x0.483	10.7	9.13	x0.459	x0.564
		run [μs/pt]					0.725	0.864	([x1.45] x1.45)	([x1.07] x1.07)	3.22	5.00	([x1.25] x1.25)	([x0.794] x0.794)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				9.75	8.74	x0.461	x0.514
		run [μs/pt]									5.28	0.761	([x1.37] x1.37)	([x1.68] x1.68)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				NOT APPLICABLE			
		run [μs/pt]												

	n=1					n=2					n=3				
summary: generation	x0.492	x0.512	x0.502	x0.502	**x0.502**	x0.452	x0.524	x0.481	x0.480	**x0.479**	x0.439	x0.583	x0.470	x0.487	**x0.486**
summary: run single-flow, hel. sum	x2.23	x2.59	x2.41	x2.41	**x2.39**	x1.45	x2.37	x1.81	x1.87	**x1.77**	x1.14	x2.04	x1.49	x1.56	**x1.42**
summary: run all-flows, single hel.	x2.73	x2.78	x2.76	x2.76	**x2.76**	x1.07	x2.40	x2.04	x1.98	**x1.79**	x0.794	x1.77	x1.59	x1.54	**x1.25**

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 2 of 3)

ID	base process	metric	n=4				n=5				n=6			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	9.62	8.33	x0.473	x0.818	10.5	9.44	x0.549	x3.35	20.5	17.9	x0.385	x28.8
		run [μs/pt]	4.97	2.45	([x1.24] x1.24)	([x1.33] x1.33)	10.8	10.8	([x1.52] x1.52)	([x1.60] x1.60)	50.4	66.0	([x0.864] x0.863)	([x2.68] x2.68)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	10.2	9.83	x0.437	x0.608	10.3	10.7	x0.495	x2.16	13.4	19.2	x0.603	x17.7
		run [μs/pt]	4.06	2.53	([x1.01] x1.01)	([x1.20] x1.20)	9.53	11.3	([x1.20] x1.20)	([x1.28] x1.27)	18.3	69.6	([x1.80] x1.80)	([x2.05] x2.05)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.88	8.30	x0.508	x0.516	9.10	8.37	x0.517	x0.723	10.5	10.3	x0.510	x2.57
		run [μs/pt]	1.52	0.880	([x1.42] x1.42)	([x1.31] x1.31)	4.62	2.87	([x1.18] x1.18)	([x1.09] x1.09)	9.71	12.4	([x1.48] x1.48)	([x1.27] x1.26)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.86	8.33	x0.494	x0.508	8.63	9.08	x0.537	x0.575	9.42	9.55	x0.534	x1.69
		run [μs/pt]	0.727	0.797	([x1.62] x1.62)	([x1.29] x1.29)	2.34	3.43	([x1.28] x1.28)	([x0.753] x0.753)	7.92	11.3	([x1.11] x1.11)	([x1.02] x1.02)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	9.43	8.22	x0.559	x0.645	11.2	8.79	x0.563	x1.68	15.7	12.0	x0.557	x9.98
		run [μs/pt]	6.51	1.39	([x1.29] x1.29)	([x1.40] x1.40)	15.5	5.12	([x1.37] x1.37)	([x1.26] x1.26)	36.3	21.3	([x1.39] x1.39)	([x1.77] x1.76)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	20.8	9.07	x0.453	x1.60	74.3	13.7	x0.515	x15.3	241	59.9	x0.441	x164
		run [μs/pt]	9.54	8.00	([x1.21] x1.21)	([x3.99] x3.98)	18.8	66.7	([x1.85] x1.85)	([x1.90] x1.90)	62.2	487	([x1.57] x1.57)	([x5.06] x5.05)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	10.7	8.93	x0.475	x0.659	17.3	9.93	x0.446	x2.37	43.5	22.7	x0.366	x15.9
		run [μs/pt]	3.86	3.21	([x1.24] x1.24)	([x1.32] x1.31)	8.67	14.4	([x1.58] x1.58)	([x1.61] x1.61)	22.1	107	([x1.69] x1.69)	([x2.37] x2.37)
8	$gg \to gg + (n-2)g$	gen. [s]	18.1	11.4	x0.447	x1.96	52.4	37.3	x0.369	x11.1	134	319	x0.386	>5H
		run [μs/pt]	7.11	20.4	([x1.87] x1.87)	([x1.40] x1.40)	19.6	172	([x2.00] x2.00)	([x1.81] x1.80)	52.6	2.99e+03	([x2.02] x2.02)	>5H
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	10.6	8.36	x0.515	x0.661	13.9	8.58	x0.551	x1.31	21.4	10.8	x0.616	x5.01
		run [μs/pt]	13.0	1.39	([x1.42] x1.42)	([x1.37] x1.37)	29.7	3.33	([x1.28] x1.28)	([x1.22] x1.22)	70.7	10.9	([x1.66] x1.65)	([x1.34] x1.34)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	9.37	9.17	x0.452	x0.500	9.00	8.32	x0.533	x0.583	10.5	8.83	x0.504	x0.678
		run [μs/pt]	2.97	1.07	([x1.13] x1.13)	([x1.17] x1.17)	5.34	1.60	([x1.28] x1.28)	([x1.16] x1.16)	10.5	3.30	([x1.37] x1.37)	([x0.890] x0.889)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.30	8.77	x0.541	x0.547	14.9	8.95	x0.477	x0.737	30.3	9.97	x0.465	x2.68
		run [μs/pt]	4.64	1.43	([x1.08] x1.08)	([x1.27] x1.27)	11.9	3.99	([x1.21] x1.21)	([x1.14] x1.13)	30.9	14.4	([x1.16] x1.16)	([x2.47] x2.47)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	9.17	8.90	x0.487	x0.502	9.59	8.39	x0.513	x0.545	10.1	8.82	x0.543	x0.715
		run [μs/pt]	3.11	1.44	([x1.23] x1.23)	([x0.948] x0.948)	5.66	1.98	([x1.31] x1.31)	([x0.954] x0.953)	11.0	3.85	([x1.21] x1.21)	([x0.828] x0.827)
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	9.26	8.64	x0.468	x0.503	12.5	9.72	x0.502	x0.683	43.6	14.0	x0.418	x1.85
		run [μs/pt]	1.06	1.92	([x1.47] x1.47)	([x0.883] x0.883)	2.38	7.90	([x1.98] x1.97)	([x0.765] x0.764)	5.83	46.2	([x2.89] x2.89)	([x0.679] x0.679)
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				17.4	11.8	x0.659	x0.939
		run [μs/pt]									7.05	22.0	([x1.51] x1.51)	([x0.633] x0.633)
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	9.42	9.18	x0.570	x0.520	12.2	10.4	x0.509	x0.741	40.8	17.9	x0.362	x2.41
		run [μs/pt]	1.77	3.20	([x0.907] x0.906)	([x0.750] x0.749)	3.04	13.5	([x1.22] x1.22)	([x0.684] x0.684)	6.92	92.0	([x1.37] x1.37)	([x0.574] x0.574)

	n=4					n=5					n=6				
summary: generation	x0.437	x0.570	x0.481	x0.491	**x0.486**	x0.369	x0.563	x0.514	x0.505	**x0.484**	x0.362	x0.659	x0.504	x0.490	**x0.440**
summary: run single-flow, hel. sum	x0.906	x1.87	x1.24	x1.30	**x1.32**	x1.18	x2.00	x1.29	x1.45	**x1.49**	x0.863	x2.89	x1.48	x1.54	**x1.51**
summary: run all-flows, single hel.	x0.749	x3.98	x1.30	x1.40	**x1.71**	x0.684	x1.90	x1.19	x1.23	**x1.67**	x0.574	x5.05	x1.30	x1.69	**x3.36**

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 3 of 3)

ID	base process	metric	n=7				n=8				n=9			
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	27.1	113	x0.484	>5H	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	68.1	804	([x1.64] x1.64)	>5H	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	24.7	123	x0.527	>5H	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	58.7	890	([x1.42] x1.42)	>5H	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	13.8	19.4	x0.573	x20.1	25.5	130	N/A	N/A	104	3.79e+03	N/A	N/A
		run [μs/pt]	23.5	74.1	([x1.65] x1.65)	([x1.93] x1.93)	65.6	692	N/A	N/A	155	2.65e+04	N/A	N/A
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	11.9	19.0	x0.537	x10.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	14.9	69.5	([x1.70] x1.70)	([x1.23] x1.23)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	26.8	36.3	x0.593	x87.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	86.9	141	([x1.53] x1.53)	([x2.98] x2.98)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	847	822	x0.511	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	144	1.25e+04	([x1.95] x1.95)	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	126	178	x0.327	>5H	383	4.69e+03	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	64.7	1.85e+03	([x1.85] x1.85)	>5H	173	7.12e+04	N/A	N/A	N/A	N/A	N/A	N/A
8	$gg \to gg + (n-2)g$	gen. [s]	780	6.59e+03	x0.253	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	145	1.24e+05	([x2.06] x2.04)	>100GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	38.3	21.5	x0.678	x32.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	168	43.7	([x1.30] x1.30)	([x2.49] x2.48)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	13.2	9.59	x0.515	x2.05	18.2	15.1	N/A	N/A	30.5	64.3	N/A	N/A
		run [μs/pt]	23.7	9.89	([x1.29] x1.29)	([x0.841] x0.840)	49.8	38.0	N/A	N/A	118	245	N/A	N/A
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	94.6	18.0	x0.410	x13.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μs/pt]	84.1	75.4	([x1.17] x1.17)	([x1.46] x1.46)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.4	10.6	x0.651	x1.66	17.5	18.2	N/A	N/A	31.9	94.6	N/A	N/A
		run [μs/pt]	23.1	11.5	([x1.30] x1.30)	([x0.797] x0.796)	47.7	41.0	N/A	N/A	113	273	N/A	N/A
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	180	59.8	x0.303	x6.34	126	1.19e+03	N/A	N/A	961	N/A	N/A	N/A
		run [μs/pt]	22.2	475	([x1.43] x1.43)	([x0.573] x0.573)	50.6	1.41e+04	N/A	N/A	220	N/A	N/A	N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	100	29.8	x0.695	x2.32	592	287	N/A	N/A	ERROR	BLOCKED DEP.	N/A	N/A
		run [μs/pt]	10.6	175	([x4.29] x4.28)	([x0.484] x0.484)	99.0	2.49e+03	N/A	N/A	ERROR	BLOCKED DEP.	N/A	N/A
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	158	119	x0.315	x7.24	143	3.45e+03	N/A	N/A	951	ERROR	N/A	N/A
		run [μs/pt]	17.6	1.41e+03	([x1.52] x1.52)	([x0.330] x0.330)	75.5	5.11e+04	N/A	N/A	219	ERROR	N/A	N/A

	n=7					n=8	n=9
summary: generation	x0.253	x0.695	x0.515	x0.491	**x0.400**	N/A	N/A
summary: run single-flow, hel. sum	x1.17	x4.28	x1.53	x1.74	**x1.64**	N/A	N/A
summary: run all-flows, single hel.	x0.330	x2.98	x1.04	x1.31	**x0.688**	N/A	N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.14 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 1 of 2)

ID	base process	metric	n=1	n=2	n=3
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.24 x0.497	8.28 x0.499	9.18 x0.459
		run [μs/pt]	0.267 ([x1.49] x1.49)	0.732 ([x1.24] x1.24)	3.23 ([x1.01] x1.01)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.39 x0.534	8.35 x0.482	9.01 x0.493
		run [μs/pt]	0.193 ([x1.90] x1.90)	0.491 ([x1.45] x1.45)	1.82 ([x1.58] x1.58)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE	8.20 x0.498	8.28 x0.513
		run [μs/pt]		0.341 ([x1.54] x1.54)	0.731 ([x1.34] x1.34)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE	8.47 x0.521	8.77 x0.462
		run [μs/pt]		0.170 ([x2.31] x2.31)	0.300 ([x2.08] x2.08)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE	8.87 x0.448	9.13 x0.488
		run [μs/pt]		1.01 ([x1.14] x1.14)	2.77 ([x1.10] x1.10)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	9.64 x0.487	10.4 x0.586
		run [μs/pt]		1.69 ([x0.890] x0.889)	14.3 ([x0.743] x0.742)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE	8.88 x0.489	8.82 x0.525
		run [μs/pt]		0.711 ([x1.09] x1.09)	3.90 ([x0.910] x0.910)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE	9.16 x0.445	11.6 x0.460
		run [μs/pt]		5.01 ([x0.888] x0.888)	56.0 ([x1.42] x1.42)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	9.27 x0.484
		run [μs/pt]			6.17 ([x0.950] x0.949)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
		run [μs/pt]			

	n=1	n=2	n=3
summary: generation	x0.497 | x0.534 | x0.516 | x0.516 | $\boxed{\text{x0.516}}$	x0.445 | x0.521 | x0.488 | x0.484 | $\boxed{\text{x0.483}}$	x0.459 | x0.586 | x0.488 | x0.497 | $\boxed{\text{x0.497}}$
summary: wall	x1.49 | x1.90 | x1.70 | x1.70 | $\boxed{\text{x1.66}}$	x0.888 | x2.31 | x1.19 | x1.32 | $\boxed{\text{x1.03}}$	x0.742 | x2.08 | x1.10 | x1.24 | $\boxed{\text{x1.24}}$

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 2 of 2)

ID	base process	metric	n=4		n=5		n=6	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	12.3	x0.553	48.1	x0.629	NOT RUN	
		run [μs/pt]	23.7	([x0.973] x0.973)	285	([x0.879] x0.879)		
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	14.4	x0.393	36.3	x0.683	NOT RUN	
		run [μs/pt]	14.6	([x1.03] x1.03)	152	([x1.18] x1.18)		
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.60	x0.530	12.2	x0.547	NOT RUN	
		run [μs/pt]	2.84	([x1.08] x1.08)	19.3	([x1.06] x1.06)		
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.99	x0.464	10.0	x0.512	NOT RUN	
		run [μs/pt]	0.996	([x1.59] x1.59)	8.41	([x1.20] x1.20)		
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.7	x0.488	25.1	x0.564	NOT RUN	
		run [μs/pt]	12.7	([x1.05] x1.05)	100	([x0.970] x0.969)		
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	27.9	x0.731	266	x2.72	NOT RUN	
		run [μs/pt]	165	([x0.756] x0.755)	9.19e+03	([x0.267] x0.267)		
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	13.0	x0.593	53.1	x0.734	NOT RUN	
		run [μs/pt]	30.1	([x1.01] x1.01)	548	([x0.760] x0.760)		
8	$gg \to gg + (n-2)g$	gen. [s]	20.0	x0.458	61.9	x0.660	NOT RUN	
		run [μs/pt]	1.10e+03	([x1.35] x1.35)	2.15e+04	([x1.34] x1.34)		
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	11.2	x0.499	18.8	x0.554	NOT RUN	
		run [μs/pt]	13.4	([x1.17] x1.17)	59.6	([x1.12] x1.12)		
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	9.16	x0.468	9.44	x0.484	NOT RUN	
		run [μs/pt]	2.96	([x1.02] x1.02)	5.31	([x1.09] x1.08)		
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.34	x0.532	13.2	x0.580	NOT RUN	
		run [μs/pt]	7.63	([x1.02] x1.02)	40.1	([x1.06] x1.06)		
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	8.60	x0.526	9.00	x0.515	NOT RUN	
		run [μs/pt]	3.14	([x1.11] x1.11)	5.71	([x1.15] x1.15)		
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	gen. [s]	9.59	x0.487	14.4	x0.607	66.5	x1.02
		run [μs/pt]	3.59	([x1.02] x1.02)	39.0	([x0.791] x0.790)	431	([x1.03] x1.03)
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		31.0	x0.892
		run [μs/pt]					76.6	([x1.42] x1.42)
15	$d\bar{d} \to u\bar{u}\,u\bar{u} + (n-4)g$	gen. [s]	9.47	x0.527	20.6	x0.577	123	x0.809
		run [μs/pt]	6.09	([x0.914] x0.914)	52.5	([x0.930] x0.930)	1.05e+03	([x0.780] x0.780)

	n=4					n=5					n=6				
summary: generation	x0.393	x0.731	x0.513	x0.518	**x0.535**	x0.484	x2.72	x0.578	x0.740	**x1.56**	x0.809	x1.02	x0.892	x0.907	**x0.884**
summary: wall	x0.755	x1.59	x1.03	x1.08	**x1.25**	x0.267	x1.34	x1.06	x0.985	**x1.01**	x0.780	x1.42	x1.03	x1.08	**x0.880**

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.15 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 1 of 2)

ID	base process	metric	n=1		n=2		n=3	
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	8.12	x0.498	8.20	x0.487	8.99	x0.501
		run [μs/pt]	0.265	([x1.49] x1.49)	0.715	([x1.27] x1.27)	3.36	([x0.979] x0.978)
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	8.18	x0.488	8.21	x0.548	9.46	x0.519
		run [μs/pt]	0.196	([x1.84] x1.83)	0.475	([x1.51] x1.51)	1.83	([x1.47] x1.47)
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE		8.92	x0.492	9.11	x0.485
		run [μs/pt]			0.339	([x1.52] x1.52)	0.727	([x1.32] x1.32)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE		9.14	x0.438	9.16	x0.450
		run [μs/pt]			0.163	([x2.40] x2.40)	0.372	([x1.66] x1.66)
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE		8.31	x0.511	9.17	x0.460
		run [μs/pt]			1.09	([x1.06] x1.06)	2.67	([x1.16] x1.16)
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE		9.06	x0.492	10.6	x0.495
		run [μs/pt]			1.80	([x0.846] x0.846)	17.2	([x0.626] x0.625)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE		8.52	x0.487	9.55	x0.481
		run [μs/pt]			0.718	([x1.08] x1.08)	3.90	([x0.905] x0.904)
8	$gg \to gg + (n-2)g$	gen. [s]	NOT APPLICABLE		9.20	x0.444	11.4	x0.441
		run [μs/pt]			5.19	([x0.841] x0.840)	57.8	([x1.55] x1.55)
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		9.62	x0.476
		run [μs/pt]					5.73	([x1.04] x1.04)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE		NOT APPLICABLE		NOT APPLICABLE	
		run [μs/pt]						

	n=1					n=2					n=3				
summary: generation	x0.488	x0.498	x0.493	x0.493	$\boxed{\text{x0.493}}$	x0.438	x0.548	x0.489	x0.487	$\boxed{\text{x0.486}}$	x0.441	x0.519	x0.481	x0.479	$\boxed{\text{x0.478}}$
summary: wall	x1.49	x1.83	x1.66	x1.66	$\boxed{\text{x1.64}}$	x0.840	x2.40	x1.17	x1.32	$\boxed{\text{x0.986}}$	x0.625	x1.66	x1.16	x1.19	$\boxed{\text{x1.29}}$

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 2 of 2)

ID	base process	metric	n=4	n=5	n=6
1	$d\bar{d} \to Z + (n-1)g$	gen. [s]	12.6 x0.529	48.4 x0.639	NOT RUN
		run [μs/pt]	26.7 ([x0.880] x0.880)	284 ([x0.932] x0.931)	
2	$u\bar{d} \to W^+ + (n-1)g$	gen. [s]	10.8 x0.738	33.2 x0.650	NOT RUN
		run [μs/pt]	14.8 ([x1.24] x1.24)	151 ([x1.19] x1.19)	
3	$d\bar{d} \to e^+e^- + (n-2)g$	gen. [s]	8.81 x0.522	11.8 x0.508	NOT RUN
		run [μs/pt]	2.73 ([x1.09] x1.09)	21.2 ([x0.933] x0.932)	
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	gen. [s]	8.94 x0.471	9.47 x0.542	NOT RUN
		run [μs/pt]	1.00 ([x1.52] x1.52)	8.91 ([x1.07] x1.07)	
5	$d\bar{d} \to ZZ + (n-2)g$	gen. [s]	10.5 x0.543	24.4 x0.575	NOT RUN
		run [μs/pt]	13.1 ([x1.03] x1.03)	108 ([x0.898] x0.897)	
6	$gg \to t\bar{t} + (n-2)g$	gen. [s]	29.8 x0.678	268 x3.91	NOT RUN
		run [μs/pt]	187 ([x0.725] x0.725)	1.00e+04 ([x0.313] x0.313)	
7	$d\bar{d} \to t\bar{t} + (n-2)g$	gen. [s]	13.3 x0.570	53.1 x0.745	NOT RUN
		run [μs/pt]	29.1 ([x1.07] x1.06)	597 ([x0.763] x0.762)	
8	$gg \to gg + (n-2)g$	gen. [s]	20.2 x0.472	69.9 x0.912	NOT RUN
		run [μs/pt]	1.27e+03 ([x1.88] x1.88)	3.60e+04 ([x3.26] x3.26)	
9	$d\bar{d} \to ZZZ + (n-3)g$	gen. [s]	11.1 x0.486	18.7 x0.557	NOT RUN
		run [μs/pt]	13.2 ([x1.19] x1.18)	59.0 ([x1.14] x1.14)	
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	gen. [s]	8.71 x0.499	9.40 x0.516	NOT RUN
		run [μs/pt]	2.90 ([x1.03] x1.03)	6.18 ([x0.937] x0.937)	
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	gen. [s]	9.16 x0.540	13.6 x0.574	NOT RUN
		run [μs/pt]	8.00 ([x0.969] x0.969)	39.4 ([x1.05] x1.05)	
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	gen. [s]	8.87 x0.485	9.65 x0.521	NOT RUN
		run [μs/pt]	3.08 ([x1.14] x1.14)	5.59 ([x1.20] x1.20)	
13	$d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	gen. [s]	9.47 x0.499	14.4 x0.656	67.9 x0.893
		run [μs/pt]	3.52 ([x1.06] x1.06)	30.5 ([x0.999] x0.998)	434 ([x1.03] x1.03)
14	$d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE	NOT APPLICABLE	30.8 x0.801
		run [μs/pt]			77.2 ([x1.33] x1.32)
15	$d\bar{d} \to u\bar{u}\, u\bar{u} + (n-4)g$	gen. [s]	10.7 x0.513	20.7 x0.567	130 x0.774
		run [μs/pt]	7.48 ([x0.908] x0.908)	52.1 ([x0.941] x0.941)	1.17e+03 ([x0.711] x0.710)

	n=4	n=5	n=6
summary: generation	x0.471 | x0.738 | x0.517 | x0.539 | **x0.553**	x0.508 | x3.91 | x0.574 | x0.848 | **x2.12**	x0.774 | x0.893 | x0.801 | x0.823 | **x0.813**
summary: wall	x0.725 | x1.88 | x1.06 | x1.12 | **x1.68**	x0.313 | x3.26 | x0.969 | x1.12 | **x2.57**	x0.710 | x1.32 | x1.03 | x1.02 | **x0.822**

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a multiplicity for which the process family does not exist. Not run marks an otherwise defined process that was intentionally excluded from the declared measurement scope. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

6 Z-plus-Jets Evaluator Ladders

The dedicated $d\bar{d} \to Z + (n-1)g$ ladders compare the independent reference with compiled JIT O1, compiled JIT O3, ASM O3, C++ O3, eager-DAG JIT O2, and recurrence JIT O2 evaluator configurations. They retain final-state multiplicities $n = 1, \ldots, 9$ for both the built-in model and the UFO-SM result set. Each row states its evaluator backend and optimization level explicitly; all other workload settings follow the benchmark contract above.

The eager and recurrence rows start from the corresponding portable JIT O2 prepared model. Each layout retains complete flow and helicity coverage; the selected-flow workload sums helicities, while the all-flow workload selects one helicity at runtime. Prepared-model creation is excluded from process-generation time. Every displayed Z-ladder timing ratio uses the matching original `AmpliCol` workload as its denominator, while every non-recurrence `pyAmpliCol` result is compared directly with the matching recurrence result at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are likewise compared directly.

Runtime comparisons between these backends also include their vectorization strategy. The JIT rows can use SIMD across phase-space points, whereas the generated ASM and C++ rows are scalar. Their ratios therefore include both code-generation quality and vectorization; they are not comparisons of otherwise identical kernels.

Worked $d\bar{d} \to Zg$ example

For $n = 2$, the explicit process is

$$d(p_1)\,\bar{d}(p_2) \to Z(p_3)\,g(p_4).$$

The report labels source particles in process order. The selected leading-colour reference order for the selected-flow timing is therefore $(2, 4, 1, 3)$: the colour line enters through $\bar{d}$, traverses the gluon, and exits through d, while the colour-singlet Z remains in the public ordering only to keep source labels unambiguous. The corresponding coloured word is $(2, 4, 1)$.

`pyAmpliCol` builds a shared-current representation for the selected process. Source nodes carry particle, momentum, helicity, flavour, and colour-flow labels; composite nodes are identified by their complete quantum state. The selected-flow workload uses the traversal $(2, 4, 1)$ and sums helicities; the all-flow workload sums every LC flow for one selected helicity. Both layouts retain complete physical coverage and are validated against the same reference point before timing.

6.1 Built-in SM dedicated $d\bar{d} \to Z+$ gluon performance

Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μs/pt]	eval [μs/pt]	gen [s]	wall [μs/pt]	eval [μs/pt]
1 `AmpliCol` reference	7.14123	0.0561071	NOT EXPOSED	0.000973	0.122144	NOT EXPOSED
1 compiled JIT O1	4.07411 (x0.571)	0.341446 (x6.09)	0.341311 (x6.08)	3.76524	0.177041 (x1.45)	0.177004 (x1.45)
1 compiled ASM O3	27.6333 (x3.87)	0.361607 (x6.44)	0.361484 (x6.44)	20.1455	0.198305 (x1.62)	0.198264 (x1.62)
1 compiled C++ O3	27.6883 (x3.88)	0.359537 (x6.41)	0.359427 (x6.41)	19.5345	0.191728 (x1.57)	0.191686 (x1.57)
1 compiled JIT O3	4.19433 (x0.587)	0.342371 (x6.1)	0.342279 (x6.1)	3.84981	0.174188 (x1.43)	0.174142 (x1.43)
1 eager-DAG JIT O2	4.02902 (x0.564)	0.547865 (x9.76)	0.54765 (x9.76)	4.59421	0.385966 (x3.16)	0.385814 (x3.16)
1 recurrence JIT O2	8.72295 (x1.22)	0.252528 (x4.5)	0.252364 (x4.5)	8.11848	0.137919 (x1.13)	0.137831 (x1.13)
2 `AmpliCol` reference	7.14717	0.361633	NOT EXPOSED	0.001072	0.280215	NOT EXPOSED
2 compiled JIT O1	4.57184 (x0.64)	0.698687 (x1.93)	0.698432 (x1.93)	4.26132	0.30645 (x1.09)	0.306302 (x1.09)
2 compiled ASM O3	41.7025 (x5.83)	0.831184 (x2.3)	0.83088 (x2.3)	30.0033	0.290057 (x1.04)	0.289919 (x1.03)
2 compiled C++ O3	41.3965 (x5.79)	0.841549 (x2.33)	0.841044 (x2.33)	29.7996	0.292342 (x1.04)	0.292228 (x1.04)
2 compiled JIT O3	4.63315 (x0.648)	0.705106 (x1.95)	0.704825 (x1.95)	4.23051	0.339985 (x1.21)	0.339808 (x1.21)
2 eager-DAG JIT O2	4.22227 (x0.591)	1.21278 (x3.35)	1.21239 (x3.35)	3.99422	0.624997 (x2.23)	0.624845 (x2.23)
2 recurrence JIT O2	8.57689 (x1.2)	0.664023 (x1.84)	0.663525 (x1.83)	8.46879	0.301337 (x1.08)	0.301108 (x1.07)
3 `AmpliCol` reference	7.32217	1.22704	NOT EXPOSED	0.00076	0.769796	NOT EXPOSED
3 compiled JIT O1	6.88209 (x0.94)	1.22521 (x0.999)	1.22482 (x0.998)	4.84391	0.538971 (x0.7)	0.538825 (x0.7)
3 compiled ASM O3	107.195 (x14.6)	2.05885 (x1.68)	2.05823 (x1.68)	62.4507	0.544592 (x0.707)	0.544379 (x0.707)
3 compiled C++ O3	107.038 (x14.6)	2.03857 (x1.66)	2.03799 (x1.66)	62.5483	0.569743 (x0.74)	0.56949 (x0.74)
3 compiled JIT O3	6.44421 (x0.88)	1.23538 (x1.01)	1.23507 (x1.01)	5.178	0.547666 (x0.711)	0.547419 (x0.711)
3 eager-DAG JIT O2	4.12088 (x0.563)	2.29103 (x1.87)	2.29033 (x1.87)	4.66144	1.09166 (x1.42)	1.09105 (x1.42)
3 recurrence JIT O2	8.43967 (x1.15)	1.6399 (x1.34)	1.63844 (x1.34)	8.3153	0.726706 (x0.944)	0.726058 (x0.943)

Original AmpliCol is the denominator only when successful and exactly linked by stored same-point evidence, directly or through current recurrence. Otherwise its status remains visible and successful pyAmpliCol values are absolute. Unverified diagnostics retain absolute clocks with a yellow marker, but never enter ratios or summaries. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value without a ratio because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no dedicated authenticated evaluator-total timing. Authenticated accumulated warmed evaluator totals appear in the eval column with enough precision to remain distinct from wall time. Its parenthesized multiplier uses the original-AmpliCol wall measurement as denominator because the legacy reference has no separate evaluator-total clock. Recurrence rows retain the independently measured narrower recurrence core in detailed evidence; it is not relabeled as evaluator total. Neither evaluator total nor recurrence core is derived from wall time or from the other. Older entries without authenticated total evidence remain marked not exposed; it is not a missing measurement. user cap: native C++/ASM generation is not attempted above n=6 (native-backend-generation-cap-n6-v1).

Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μs/pt]	eval [μs/pt]	gen [s]	wall [μs/pt]	eval [μs/pt]
4 `AmpliCol` reference	7.83511	4.46017	NOT EXPOSED	0.004892	3.05425	NOT EXPOSED
4 compiled JIT O1	12.0474 (x1.54)	3.16632 (x0.71)	3.16516 (x0.71)	9.28338	1.38931 (x0.455)	1.38881 (x0.455)
4 compiled ASM O3	284.576 (x36.3)	6.97794 (x1.56)	6.97596 (x1.56)	201.799	2.01364 (x0.659)	2.01294 (x0.659)
4 compiled C++ O3	283.883 (x36.2)	6.96784 (x1.56)	6.9654 (x1.56)	201.878	2.0224 (x0.662)	2.02189 (x0.662)
4 compiled JIT O3	12.2098 (x1.56)	3.20899 (x0.719)	3.20797 (x0.719)	8.90448	1.38387 (x0.453)	1.38346 (x0.453)
4 eager-DAG JIT O2	4.47437 (x0.571)	6.06552 (x1.36)	6.06375 (x1.36)	6.82557	3.34166 (x1.09)	3.3406 (x1.09)
4 recurrence JIT O2	9.96583 (x1.27)	4.92677 (x1.1)	4.92198 (x1.1)	9.28432	2.53255 (x0.829)	2.52961 (x0.828)
5 `AmpliCol` reference	8.16277	13.1733	NOT EXPOSED	0.014886	15.2814	NOT EXPOSED
5 compiled JIT O1	37.7636 (x4.63)	10.0621 (x0.764)	10.0594 (x0.764)	30.6412	6.93978 (x0.454)	6.93799 (x0.454)
5 compiled ASM O3	1133.16 (x139)	24.6947 (x1.87)	24.6874 (x1.87)	823.947	18.382 (x1.2)	18.3753 (x1.2)
5 compiled C++ O3	1135.02 (x139)	24.5503 (x1.86)	24.5432 (x1.86)	819.775	18.3861 (x1.2)	18.3809 (x1.2)
5 compiled JIT O3	38.3904 (x4.7)	9.96468 (x0.756)	9.96086 (x0.756)	30.7046	7.03598 (x0.46)	7.03414 (x0.46)
5 eager-DAG JIT O2	5.29804 (x0.649)	16.4159 (x1.25)	16.4106 (x1.25)	30.7725	16.9961 (x1.11)	16.991 (x1.11)
5 recurrence JIT O2	11.5065 (x1.41)	11.9645 (x0.908)	11.9501 (x0.907)	10.0516	11.2379 (x0.735)	11.2262 (x0.735)
6 `AmpliCol` reference	8.78326	34.0798	NOT EXPOSED	0.140064	96.2781	NOT EXPOSED
6 compiled JIT O1	246.15 (x28)	37.5796 (x1.1)	37.5653 (x1.1)	217.57	73.1599 (x0.76)	73.1312 (x0.76)
6 compiled ASM O3	7258.1 (x826)	89.9324 (x2.64)	89.9021 (x2.64)	6268.92	264.106 (x2.74)	263.91 (x2.74)
6 compiled C++ O3	7287.93 (x830)	91.0267 (x2.67)	90.9963 (x2.67)	6218.48	265.407 (x2.76)	265.203 (x2.75)
6 compiled JIT O3	229.734 (x26.2)	40.816 (x1.2)	40.8006 (x1.2)	212.308	71.6492 (x0.744)	71.6249 (x0.744)
6 eager-DAG JIT O2	7.89024 (x0.898)	41.9815 (x1.23)	41.9676 (x1.23)	515.748	180.331 (x1.87)	180.281 (x1.87)
6 recurrence JIT O2	20.4988 (x2.33)	26.6488 (x0.782)	26.6175 (x0.781)	17.8836	68.8022 (x0.715)	68.7253 (x0.714)

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Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 3 of 3)

n setup	selected flow, helicity sum gen [s]	wall [μs/pt]	eval [μs/pt]	all flows, single helicity gen [s]	wall [μs/pt]	eval [μs/pt]
7 AmpliCol reference	9.92791	90.8874	NOT EXPOSED	1.44645	839.483	NOT EXPOSED
7 compiled JIT O1	45.201 (x4.55)	114.872 (x1.26)	114.822 (x1.26)	2194.61	742.643 (x0.885)	742.272 (x0.884)
7 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled JIT O3	45.5885 (x4.59)	117.712 (x1.3)	117.653 (x1.29)	2220.39	838.795 (x0.999)	838.111 (x0.998)
7 eager-DAG JIT O2	13.1321 (x1.32)	109.205 (x1.2)	109.132 (x1.2)	>5H	>5H	>5H
7 recurrence JIT O2	27.128 (x2.73)	68.2228 (x0.751)	68.1091 (x0.749)	112.511	820.97 (x0.978)	819.714 (x0.976)
8 AmpliCol reference	N/A	N/A	NOT EXPOSED	N/A	N/A	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 AmpliCol reference	N/A	N/A	NOT EXPOSED	N/A	N/A	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A

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6.2 UFO-SM dedicated $d\bar{d} \to Z+$ gluon performance

UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 1 of 3)

n setup	selected flow, helicity sum gen [s]	wall [µs/pt]	eval [µs/pt]	all flows, single helicity gen [s]	wall [µs/pt]	eval [µs/pt]
1 `AmpliCol` reference	7.14123	0.0561071	NOT EXPOSED	0.000973	0.122144	NOT EXPOSED
1 compiled JIT O1	6.46784 (x0.906)	0.365434 (x6.51)	0.365363 (x6.51)	6.07971	0.177515 (x1.45)	0.177475 (x1.45)
1 compiled ASM O3	41.1529 (x5.76)	0.375674 (x6.7)	0.375541 (x6.69)	28.8276	0.171397 (x1.4)	0.171338 (x1.4)
1 compiled C++ O3	41.1632 (x5.76)	0.371501 (x6.62)	0.371386 (x6.62)	28.9599	0.173894 (x1.42)	0.173823 (x1.42)
1 compiled JIT O3	6.42984 (x0.9)	0.365975 (x6.52)	0.365902 (x6.52)	6.03375	0.177014 (x1.45)	0.176972 (x1.45)
1 eager-DAG JIT O2	5.92127 (x0.829)	0.546008 (x9.73)	0.545891 (x9.73)	5.90871	0.388965 (x3.18)	0.388881 (x3.18)
1 recurrence JIT O2	15.2358 (x2.13)	0.263786 (x4.7)	0.263607 (x4.7)	14.748	0.139084 (x1.14)	0.138996 (x1.14)
2 `AmpliCol` reference	7.14717	0.361633	NOT EXPOSED	0.001072	0.280215	NOT EXPOSED
2 compiled JIT O1	6.42934 (x0.9)	0.706879 (x1.95)	0.706718 (x1.95)	6.0195	0.311281 (x1.11)	0.311188 (x1.11)
2 compiled ASM O3	58.1509 (x8.14)	0.865032 (x2.39)	0.864707 (x2.39)	37.8195	0.295294 (x1.05)	0.295221 (x1.05)
2 compiled C++ O3	58.3242 (x8.16)	0.864651 (x2.39)	0.864421 (x2.39)	38.4493	0.312434 (x1.11)	0.312334 (x1.11)
2 compiled JIT O3	6.84982 (x0.958)	0.735143 (x2.03)	0.734931 (x2.03)	6.3392	0.312515 (x1.12)	0.312411 (x1.11)
2 eager-DAG JIT O2	6.00711 (x0.84)	1.23344 (x3.41)	1.23309 (x3.41)	6.55825	0.636562 (x2.27)	0.636416 (x2.27)
2 recurrence JIT O2	15.0181 (x2.1)	0.722154 (x2)	0.721617 (x2)	15.7569	0.304709 (x1.09)	0.304416 (x1.09)
3 `AmpliCol` reference	7.32217	1.22704	NOT EXPOSED	0.00076	0.769796	NOT EXPOSED
3 compiled JIT O1	9.10736 (x1.24)	1.29494 (x1.06)	1.29445 (x1.05)	6.71812	0.550451 (x0.715)	0.550305 (x0.715)
3 compiled ASM O3	125.059 (x17.1)	2.11426 (x1.72)	2.11369 (x1.72)	71.4351	0.568459 (x0.738)	0.568269 (x0.738)
3 compiled C++ O3	125.902 (x17.2)	2.13683 (x1.74)	2.13572 (x1.74)	72.1243	0.577409 (x0.75)	0.57726 (x0.75)
3 compiled JIT O3	8.75718 (x1.2)	1.30266 (x1.06)	1.30227 (x1.06)	6.71865	0.540539 (x0.702)	0.540401 (x0.702)
3 eager-DAG JIT O2	6.70714 (x0.916)	2.35213 (x1.92)	2.35155 (x1.92)	6.31544	1.11132 (x1.44)	1.11105 (x1.44)
3 recurrence JIT O2	15.5279 (x2.12)	1.74832 (x1.42)	1.74667 (x1.42)	14.6562	0.721653 (x0.937)	0.720812 (x0.936)

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UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μs/pt]	eval [μs/pt]	gen [s]	wall [μs/pt]	eval [μs/pt]
4 AmpliCol reference	7.83511	4.46017	NOT EXPOSED	0.004892	3.05425	NOT EXPOSED
4 compiled JIT O1	14.7277 (x1.88)	3.50499 (x0.786)	3.50343 (x0.785)	10.6035	1.44782 (x0.474)	1.4473 (x0.474)
4 compiled ASM O3	298.426 (x38.1)	7.51611 (x1.69)	7.51411 (x1.68)	198.66	2.10712 (x0.69)	2.10665 (x0.69)
4 compiled C++ O3	301.299 (x38.5)	7.70348 (x1.73)	7.70139 (x1.73)	200.613	2.20633 (x0.722)	2.20584 (x0.722)
4 compiled JIT O3	15.0631 (x1.92)	3.50431 (x0.786)	3.50309 (x0.785)	11.2549	1.4372 (x0.471)	1.43682 (x0.47)
4 eager-DAG JIT O2	7.20325 (x0.919)	6.12836 (x1.37)	6.12597 (x1.37)	9.35313	3.28043 (x1.07)	3.27929 (x1.07)
4 recurrence JIT O2	16.6299 (x2.12)	5.08956 (x1.14)	5.08429 (x1.14)	17.2659	2.96056 (x0.969)	2.9573 (x0.968)
5 AmpliCol reference	8.16277	13.1733	NOT EXPOSED	0.014886	15.2814	NOT EXPOSED
5 compiled JIT O1	43.1518 (x5.29)	11.4547 (x0.87)	11.4512 (x0.869)	34.7333	7.58763 (x0.497)	7.58561 (x0.496)
5 compiled ASM O3	1195.57 (x146)	25.47 (x1.93)	25.4616 (x1.93)	848.666	19.3868 (x1.27)	19.3817 (x1.27)
5 compiled C++ O3	1405.31 (x172)	25.5946 (x1.94)	25.5862 (x1.94)	884.572	19.1203 (x1.25)	19.1115 (x1.25)
5 compiled JIT O3	40.8157 (x5)	11.4751 (x0.871)	11.4718 (x0.871)	33.5504	7.78104 (x0.509)	7.77717 (x0.509)
5 eager-DAG JIT O2	7.80752 (x0.956)	16.4374 (x1.25)	16.4322 (x1.25)	35.2329	17.1785 (x1.12)	17.1739 (x1.12)
5 recurrence JIT O2	17.9766 (x2.2)	11.9186 (x0.905)	11.9036 (x0.904)	16.0439	11.4804 (x0.751)	11.4679 (x0.75)
6 AmpliCol reference	8.78326	34.0798	NOT EXPOSED	0.140064	96.2781	NOT EXPOSED
6 compiled JIT O1	243.27 (x27.7)	50.4948 (x1.48)	50.4738 (x1.48)	216.958	108.39 (x1.13)	108.355 (x1.13)
6 compiled ASM O3	8418.45 (x958)	92.47 (x2.71)	92.4399 (x2.71)	7571.67	262.048 (x2.72)	261.858 (x2.72)
6 compiled C++ O3	>5H	>5H	>5H	BLOCKED DEP.	BLOCKED DEP.	BLOCKED DEP.
6 compiled JIT O3	>5H	>5H	>5H	BLOCKED DEP.	BLOCKED DEP.	BLOCKED DEP.
6 eager-DAG JIT O2	12.1094 (x1.38)	45.4514 (x1.33)	45.4313 (x1.33)	529.771	180.729 (x1.88)	180.646 (x1.88)
6 recurrence JIT O2	21.5878 (x2.46)	27.7158 (x0.813)	27.677 (x0.812)	25.1525	75.0225 (x0.779)	74.9072 (x0.778)

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UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 3 of 3)

n setup	selected flow, helicity sum gen [s]	wall [μs/pt]	eval [μs/pt]	all flows, single helicity gen [s]	wall [μs/pt]	eval [μs/pt]
7 `AmpliCol` reference	9.92791	90.8874	NOT EXPOSED	1.44645	839.483	NOT EXPOSED
7 compiled JIT O1	55.882 (x5.63)	143.215 (x1.58)	143.141 (x1.57)	2148.46	804.66 (x0.959)	804.278 (x0.958)
7 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled JIT O3	51.1251 (x5.15)	134.691 (x1.48)	134.623 (x1.48)	2210.71	801.841 (x0.955)	801.441 (x0.955)
7 eager-DAG JIT O2	16.3508 (x1.65)	109.57 (x1.21)	109.517 (x1.2)	>5H	>5H	>5H
7 recurrence JIT O2	35.3721 (x3.56)	70.5185 (x0.776)	70.3766 (x0.774)	128.158	881.347 (x1.05)	880.04 (x1.05)
8 `AmpliCol` reference	N/A	N/A	NOT EXPOSED	N/A	N/A	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 `AmpliCol` reference	N/A	N/A	NOT EXPOSED	N/A	N/A	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A

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Each LC workload reports process-generation time in seconds and native wall time in microseconds per phase-space point, together with a separately measured execution attribution when exposed. Compiled rows use the labelled JIT, ASM, or C++ backend; eager and recurrence rows use portable JIT O2 prepared models. Parenthesized wall-time and selected-flow generation values compare each pyAmpliCol row with the corresponding AmpliCol workload. For compiled and eager rows, wall time is the authenticated warmed total-evaluator boundary; not exposed refers only to a narrower execution attribution. All-flow generation is shown absolutely without a ratio because its setup boundary differs from the reference.

7 Colour-Singlet Model Ladders

Both colour-singlet ladders use compiled JIT O3 evaluation and the full-colour contract. The scalar-contact ladder retains $n = 2, \ldots, 8$ final scalars and tests high-valence contact decomposition, identical-particle normalization, and source parity without Standard-Model-specific kernels. The scalar-gravity ladder retains $n = 2, \ldots, 4$ final gravitons and exercises spin-two wavefunctions, tensor propagators, and momentum-dependent vertices. Because every external particle in these models is a colour singlet, LC, NLC, and full colour reduce to the same unit colour contraction.

For these ladders, the relative-difference row is $\big||\mathcal{M}|^2_{\mathrm{f64}} - |\mathcal{M}|^2_{\mathrm{hp}}\big| / \max(|\mathcal{M}|^2_{\mathrm{hp}}, 10^{-300})$, where the high-precision value uses the same phase-space point. It is a numerical stability check, not a performance ratio.

7.1 Scalar-contact ladder

	scalar_0 scalar_0 > n*scalar_0						
metric	n=2	n=3	n=4	n=5	n=6	n=7	n=8
generation [s]	3.61087	4.30139	6.6474	15.1274	42.7201	160.376	N/A
wall [μs/pt]	0.172033	0.253307	0.541287	0.893346	2.0672	3.08998	N/A
evaluator total [μs/pt]	0.17199	0.253238	0.54111	0.893069	2.06663	3.08915	N/A
matrix element	0.5	0.167	0.0417	0.00833	0.00139	0.000198	N/A
relative diff. vs hp	1.11e-16	1.67e-16	0	0	0	0	N/A

Wall time is the common runtime observable. Evaluator total is the independently measured accumulated warmed evaluator clock from its dedicated authenticated record; it is never copied from or derived from wall time. Unverified diagnostics retain absolute generation, wall, and evaluator clocks with a yellow marker; they are not successes. NOT EXPOSED denotes a successful wall measurement, and older entries without dedicated evaluator-total evidence remain valid; it does not denote a missing result.

7.2 Scalar-gravity ladder

	scalar_0 scalar_0 > n*graviton		
metric	n=2	n=3	n=4
generation [s]	4.82911	298.829	2210.87
wall [μs/pt]	3.95437	4187.03	39548.4
evaluator total [μs/pt]	3.95348	4186.18	39545.8
matrix element	1.61e+09	9.46e+11	3.07e+13
relative diff. vs hp	1.04e-15	4.13e-15	1.92e-14

Wall time is the common runtime observable. Evaluator total is the independently measured accumulated warmed evaluator clock from its dedicated authenticated record; it is never copied from or derived from wall time. Unverified diagnostics retain absolute generation, wall, and evaluator clocks with a yellow marker; they are not successes. NOT EXPOSED denotes a successful wall measurement, and older entries without dedicated evaluator-total evidence remain valid; it does not denote a missing result.

8 Interpretation

Performance comparisons are meaningful only for a baseline measured with the same observables, parameters, selectors, and hardware profile. The result tables therefore present validated, architecture-local comparisons; they do not combine measurements from different machines. An N/A entry denotes only a structurally inapplicable cell or observable; every applicable cell contains a validated measurement or documented resource-limit status. Numerical agreement, timing uncertainty, and software revision remain part of the published result set even when a compact table shows only the central timing.

9 Worked $d\bar{d} \to Zgg$ Example

This example illustrates how a familiar tree-level process is represented by reusable off-shell currents. It also fixes the momentum, helicity, and colour conventions used in the performance tables.

9.1 External legs and conventions

The physical process is

$$d(p_1)\,\bar{d}(p_2) \longrightarrow Z(p_3)\,g(p_4)\,g(p_5), \qquad p_1 + p_2 = p_3 + p_4 + p_5. \tag{6}$$

The current recursion uses the all-outgoing convention

$$q_1 = -p_1, \quad q_2 = -p_2, \quad q_i = p_i \ (i = 3, 4, 5), \qquad \sum_{i=1}^{5} q_i = 0. \tag{7}$$

External momenta are supplied in the physical convention of Eq. (6) and crossed to the all-outgoing convention before the recursion is evaluated.

label	physical leg	outgoing state	momentum	states	colour role
1	incoming d	outgoing $\bar{d}$	$q_1 = -p_1$	2	$\bar{\mathbf{3}}$, line end
2	incoming $\bar{d}$	outgoing d	$q_2 = -p_2$	2	$\mathbf{3}$, line start
3	outgoing Z	outgoing Z	$q_3 = p_3$	3	singlet
4	outgoing g	outgoing g	$q_4 = p_4$	2	adjoint insertion
5	outgoing g	outgoing g	$q_5 = p_5$	2	adjoint insertion

There are $2 \times 2 \times 3 \times 2 \times 2 = 48$ physical helicity configurations. Twenty-four vanish identically because the massless quark line permits only chirality-compatible initial-state helicities; they remain explicit zero entries in resolved results.

At leading colour the Z is absent from the colour word. Complete coverage contains the two orderings

$$c_0 = (2, 4, 5, 1), \qquad c_1 = (2, 5, 4, 1). \tag{8}$$

Both are retained during generation, so either flow can be selected at evaluation time.

9.2 Reusable current recursion

For a nonempty source subset I, define its momentum by

$$q_I = \sum_{i \in I} q_i. \tag{9}$$

A one-leg current is the appropriate external spinor, polarization vector, or scalar wavefunction. Composite currents obey the binary recursion

$$J^a_{t,\Omega}(I) = {P_t}^a{}_b(q_I) \sum_{\substack{I = A \dot\cup B \\ A, B \neq \varnothing}} \sum_{v\,:\,t_A t_B \to t} \kappa_v \, {V_v}^b{}_{cd}(q_A, q_B) J^c_{t_A,\Omega_A}(A) J^d_{t_B,\Omega_B}(B). \tag{10}$$

Here t is the particle state, P_t is its propagator, V_v is an allowed interaction, and Ω collects the quantum numbers required to identify the current. Only combinations consistent with the model, perturbative order, and colour ordering enter the sum.

Representative terms along the quark line are

$$J_d(2, 5) = P_d \, V_{dgd}\big[J_d(2), J_g(5)\big], \tag{11}$$

$$J_g(4, 5) = P_g \, V_{ggg}\big[J_g(4), J_g(5)\big], \tag{12}$$

$$J_d(2, 4, 5) = P_d\Big(V_{dgd}\big[J_d(2, 5), J_g(4)\big] + V_{dgd}\big[J_d(2, 4), J_g(5)\big] + V_{dgd}\big[J_d(2), J_g(4, 5)\big]\Big), \tag{13}$$

$$J_d(2, 3, 4, 5) = P_d \sum_{A \dot\cup B = \{2,3,4,5\}} V\big[J(A), J(B)\big]. \tag{14}$$

Equation (14) includes every admissible placement of the Z emission and both gluon insertions. Ordered currents such as $J_d(2, 5, 4)$ and $J_d(2, 4, 5)$ remain distinct even though their momentum subset is the same.

The final closure joins the endpoint current to source label 1:

$$\mathcal{A}^{(r)}_{h,c} = C^{(r)}_{\dot{a}a} \, J^{\dot{a}}_{\bar{d}}(1) \, J^a_d(2, 3, 4, 5), \qquad \mathcal{A}_{h,c} = \sum_{r \in G(h,c)} \mathcal{A}^{(r)}_{h,c}. \tag{15}$$

The set $G(h, c)$ contains contributions that must be summed coherently. Identical subcurrents are evaluated once and reused wherever their complete quantum-number and ordering labels agree. This sharing is the essential difference between the current DAG and an independent evaluation of every Feynman diagram.

9.3 Resolved observable and normalization

The two leading-colour layouts described in Sec. 11 evaluate the same complete set of helicity and flow components. One is optimized for a selected flow with summed helicities; the other shares work across all flows for a selected helicity. Their resolved components and their summed matrix element are numerically identical.

For the built-in Standard Model, the normalization used in this example is

$$|\mathcal{M}|^2 = \mathcal{N} \sum_{h,c} w_{h,c} \left| \sum_{r \in G(h,c)} \mathcal{A}_{h,c}^{(r)} \right|^2, \qquad \mathcal{N} = \frac{27\,(4\pi\alpha_s)^2 (8\pi\alpha_{\mathrm{EW}})}{36 \times 2}. \tag{16}$$

The denominator averages over the incoming colour and spin states, while the factor $2 = 2!$ accounts for the identical final-state gluons. Coupling powers, colour weights, coherent groups, and symmetry factors are retained separately so that changing numerical model parameters does not alter the process definition.

10 Shared-Current DAG and Numerical Execution

The current recursion becomes a directed acyclic graph (DAG) when physically identical intermediate currents are shared. This section summarizes the identity and ordering needed for that reuse and relates them to the three execution modes.

10.1 Current identity and reuse

Two contributions share a current only when every physics-relevant label agrees. Schematically,

$$\Omega(J) = \big(t, I, H, s, F, Q, C, O, a, \pi_I\big), \tag{17}$$

where the entries identify the particle species, external subset, helicity and spin ancestry, flavour and additive charges, colour state, perturbative order, auxiliary-current type, and colour-sensitive ordering. The subset determines the current momentum $q_I = \sum_{i \in I} q_i$, while the ordered labels prevent distinct colour traversals from being merged.

Exact identity permits one stored current to serve every amplitude contribution that needs it. A second kind of reuse applies when different currents require the same numerical kernel value up to a known exact factor. Neither form of reuse merges distinct physical helicity or colour-flow components.

10.2 Topological ordering

The graph is evaluated in increasing external-subset size. Every input is therefore available before the current that consumes it. For the process in Sec. 9, the ordering is schematically

$$\boxed{\text{sources}} \longrightarrow \boxed{|I| = 2} \longrightarrow \boxed{|I| = 3} \longrightarrow \boxed{|I| = 4} \longrightarrow \boxed{\text{amplitude closures}}. \tag{18}$$

At each stage, all admissible partitions contributing to the same current are summed coherently before propagation. The final closures join the largest retained currents to the remaining external states. This ordering makes the dependency structure explicit without assigning physical meaning to a particular storage layout.

10.3 Numerical realization

Compiled mode combines many relations in a stage into process-specific multi-output evaluators. Eager mode follows the same process graph using reusable prepared-model kernels. Recurrence mode represents the current and closure sequence directly. All three preserve the same coherent sums, colour contraction, helicity treatment, symmetry weights, and normalization.

For resolved leading-colour evaluation, the public result has the form

$$R_{p,h,f} = \big|\mathcal{M}_{h,f}(p)\big|^2, \qquad \text{shape}(R) = (N_{\text{point}}, N_{\text{helicity}}, N_{\text{flow}}). \tag{19}$$

Structural zeros remain explicit, and summing the non-point axes must reproduce the directly evaluated total within the stated numerical tolerance.

10.4 Timing boundary

The primary runtime quantity is the complete native wall time after caller-language array conversion. It includes source and momentum preparation, the mode-specific numerical evaluation, selectors, closures, and final reduction. Packing performed by the caller is outside this boundary.

Some backends additionally expose a narrower execution attribution. Because its precise boundary is mode-specific, it is reported only as a supplementary quantity and only when measured directly. The complete wall time remains the common observable used for comparisons; an unavailable attribution is marked explicitly rather than inferred.

11 Reusable Leading-Colour Execution Layouts

Leading-colour coverage and its numerical layout are separate concepts. Let $\mathcal{H}$ be the retained helicities and $\mathcal{F}$ the retained physical colour flows. Complete leading-colour coverage represents

$$\mathcal{C}_{\mathrm{LC}} = \mathcal{H} \times \mathcal{F} \tag{20}$$

and can expose every resolved amplitude $\mathcal{A}_{hf}$. For selected subsets $S_H \subseteq \mathcal{H}$ and $S_F \subseteq \mathcal{F}$, the observable is

$$\mathcal{W}(S_H, S_F) = \mathcal{N} \sum_{h \in S_H} \sum_{f \in S_F} w_{hf} \left| \sum_{r \in G(h,f)} \mathcal{A}_{hf}^{(r)} \right|^2 . \tag{21}$$

Here $G(h, f)$ is a coherent contribution group, w_{hf} contains exact symmetry and leading-colour weights, and $\mathcal{N}$ is the process normalization. The layout changes how this sum is scheduled, not its components or normalization.

11.1 Two complete-coverage layouts

The report measures two layouts because selected-flow and all-flow workloads offer different opportunities for sharing:

property	**topology replay**	**all-flow union**
Measured workload	one selected flow, summed helicities	all flows, one selected helicity
Retained physics	every flow and helicity	every flow and helicity
Main reuse	equivalent flow topologies related by exact permutations and factors	currents and closures shared directly across all flows
Availability	compiled, eager, and recurrence	compiled, eager, and recurrence

In topology replay, exact relations partition physical flows into equivalence classes. If $\rho(f)$ represents the class containing f, the calculation records a permutation π_f, an exact factor s_f, and the corresponding weight:

$$\mathcal{A}_{hf}(p) = s_f \, \mathcal{A}_{\pi_f(h),\rho(f)}\big(\pi_f(p)\big) . \tag{22}$$

This avoids constructing an independent current graph for every physical ordering and is suited to selected-flow, helicity-summed evaluation.

The all-flow union instead merges the dependency graphs of all resolved components:

$$D_{\mathrm{union}} = \left(\bigcup_{h \in \mathcal{H},\, f \in \mathcal{F}} D_{hf} \right) / \!\sim, \tag{23}$$

where $\sim$ identifies currents, kernel values, and closures that are algebraically equal up to retained exact factors. For a selected helicity, one graph traversal then reuses intermediate currents across all flows.

Compiled mode specializes the chosen layout to process-wide stage evaluators. Eager mode applies prepared-model kernels to the same dependency structure, and recurrence mode constructs a compact current schedule directly. These execution choices do not change $\mathcal{H}$, $\mathcal{F}$, or Eq. (21).

11.2 Runtime selection

Flow and helicity selections refer to physical components. A selection may be applied to a complete batch or specified independently for each point; omitting an axis sums all retained entries on that axis. Rows with equal selections can be grouped into contiguous numerical batches. This grouping changes neither the generated coverage nor the resolved result.

11.3 Generation and reference boundaries

The two layouts build different graphs and numerical schedules, so their process-generation times are distinct observables:

$$T_{\mathrm{gen}}^{\mathrm{replay}} \quad \text{and} \quad T_{\mathrm{gen}}^{\mathrm{union}}. \tag{24}$$

Cross-mode comparisons use the same layout on both sides. Preparation of reusable model-level kernels occurs before the process-generation timer and is not included in these table values.

The original-`AmpliCol` reference uses two different routes. For the selected-flow workload, its generation value measures creation of the process library. For the all-flow workload, the reported setup value measures preparation of direct all-flow evaluation. The corresponding runtime measurements implement the same physical workloads as **`pyAmpliCol`**, but the all-flow setup quantities do not define a like-for-like process-generation speedup. The table therefore reports both quantities absolutely and marks their generation comparison as not comparable.

11.4 NLC and full colour

The alternative layouts apply only at leading colour. NLC and full-colour evaluation construct an amplitude basis and contract it with the appropriate colour metric:

$$\mathcal{W}_{\text{contracted}} = \mathcal{N} \sum_{h} \sum_{a,b} \mathcal{A}^{*}_{ha}\, C_{ab}\, \mathcal{A}_{hb}. \tag{25}$$

These treatments do not expose the physical leading-colour flow axis, so there is no topology-replay or all-flow-union distinction.

12 Compiled, Eager, and Recurrence Execution

The three execution modes differ in how they turn the same current recursion into numerical work. Compiled mode specializes groups of current relations to one process. Eager mode applies reusable model kernels according to a process-specific schedule. Recurrence mode constructs a compact schedule of currents, contributions, and closures directly. These choices change the balance between process-generation cost and runtime work, but not the observable being evaluated.

Recurrence JIT O2 is the default execution mode.

mode	process preparation	evaluation	characteristic balance
Compiled	Builds process-specific stage evaluators	Evaluates a small number of process-wide, multi-output stages	More preparation; greater fusion across a process
Eager	Maps the process graph to reusable prepared-model kernels	Applies those kernels according to the process dependencies	Reuses model work across processes
Recurrence	Builds a compact schedule directly from the current recursion	Evaluates scheduled currents, contributions, and closures	Avoids construction of a generic process DAG

12.1 Common physics contract

All three modes retain the same external states, perturbative orders, helicities, colour components, coherent amplitude groups, symmetry factors, and initial-state averages. In a contracted colour basis their common observable has the schematic form

$$\mathcal{M}_p = \mathcal{N} \sum_h \sum_{a,b} A_{ha}(p)^* \, C_{ab} \, A_{hb}(p), \tag{26}$$

with the corresponding resolved form retained when helicity or leading-colour flow components are requested. Each mode must reproduce both the directly evaluated total and the sum of those resolved components.

Current contributions are accumulated coherently before a propagator or other current transformation is applied. Final closures then form the amplitude components entering Eq. (26). This ordering is common to the three modes and prevents execution strategy from changing the normalization.

12.2 Prepared-model boundary

A prepared model contains reusable numerical information derived from the model's particles, parameters, propagators, and interaction tensors. Eager and recurrence JIT O2 measurements reuse this model-level preparation and add only the process-dependent schedule. Compiled JIT O3 measurements instead include construction and compilation of their process-specific evaluators.

Prepared-model creation occurs outside the process-generation timer. The reported eager and recurrence generation times therefore describe adding a process to an already prepared model, not importing and compiling the model itself.

12.3 Batches, selectors, and portability

Evaluation is batched over phase-space points, allowing the JIT backend to use the point dimension for vectorization. Partitioning a large input batch changes neither the process representation nor the numerical result.

Helicity and colour-flow selectors refer to physical resolved components. A complete leading-colour calculation retains every declared helicity and flow, so selecting one flow with summed helicities or one helicity with all flows does not alter the underlying physics coverage. The two layouts used for these workloads are described in Sec. 11.

Portable JIT state is compiled for the receiving processor when loaded. Target-native ASM and C++ payloads remain architecture-specific, and the exact symbolic representation is retained where higher-precision evaluation is supported.

12.4 Interpretation

Compiled mode may benefit most when process-wide fusion outweighs its additional preparation cost. Eager mode can reduce repeated preparation when many processes share a model, while recurrence provides a compact direct representation of the current construction. The tables compare these trade-offs under identical physics, selector, precision, and validation conditions; no mode is assumed to be universally fastest.

13 Supported UFO and Serialized-JSON Models

`pyAmpliCol` accepts a documented subset of the Universal FeynRules Output (UFO) format and an equivalent serialized-JSON representation. Both sources are normalized before process generation and then use the same current recursion, colour treatment, and numerical interfaces as the built-in models. Unsupported structures are rejected explicitly rather than interpreted approximately.

13.1 Sources and normalization

The two input routes are

$$\{\text{trusted UFO module, serialized JSON}\} \longrightarrow \text{normalized model} \longrightarrow \text{numerical kernels.} \tag{27}$$

A UFO module is executable Python and must therefore come from a trusted source. The serialized JSON form is data-only and is preferable for portable exchange and archival. After loading, both forms describe the same particles, parameters, propagators, interactions, coupling orders, Lorentz tensors, and colour tensors.

Model symbols remain in a model-local namespace, so equally named parameters from different models are not identified accidentally. Normalization also makes component orderings and particle conjugation explicit. A model is accepted for ordinary process generation only when the normalized model satisfies the support boundary below; an unrepresentable requested interaction is rejected explicitly.

13.2 Interactions and exact verification

A UFO vertex is a sum over colour and Lorentz structures,

$$\mathcal{V}(1,\ldots,n) = \sum_{a,b} g_{ab}\, C_a(c_1,\ldots,c_n)\, L_b(s_1,p_1;\ldots;s_n,p_n). \tag{28}$$

Each term retains its coupling-order assignment. The supported Lorentz basis includes scalars, momenta, metrics, Dirac identity and gamma matrices, γ_5, chiral projectors, slashed momenta, and sigma tensors. The supported SU(3) representations are the singlet, fundamental, antifundamental, and adjoint.

Colour projection and symmetry reuse are based on exact normalized expressions. Numerical agreement at a sample point is not used to establish an algebraic identity. When an identity needed only for reuse cannot be proved, the corresponding contributions are evaluated separately. When an identity is required to represent an interaction at all, failure to establish it causes the model or process to be rejected.

Leading-colour singlet subtraction, for example, follows from the fundamental Fierz identity

$$(T^a)_i{}^{\bar{\jmath}}(T^a)_k{}^{\bar{l}} = T_R\left(\delta_i{}^{\bar{l}}\delta_k{}^{\bar{\jmath}} - \frac{1}{N_c}\delta_i{}^{\bar{\jmath}}\delta_k{}^{\bar{l}}\right). \tag{29}$$

It is used only when the colour representation, Lorentz interaction, and vector propagator satisfy the corresponding exact conditions. The same principle applies to vectorlike parity, Yang–Mills relations, and current reflection.

13.3 Contact interactions and propagators

Supported contact interactions are rewritten with non-propagating auxiliary currents only when the original tensor can be reconstructed exactly. The physical coupling appears once, and the rewriting introduces no artificial pole. Colour-singlet scalar contacts provide the validation ladder for this construction. Higher-valence singlet interactions are accepted when their normalized tensor admits the same exact reconstruction.

A coloured four-vector gauge contact is accepted only for the two-f structure

$$C^{a_1a_2a_3a_4} = c\, f^{a_ia_jb} f^{ba_ka_r}, \tag{30}$$

up to exact permutations, with one shared adjoint index and a matching Lorentz factorization. Other coloured contact tensors require a separately supported intermediate colour channel.

Standard scalar, Dirac, vector, and spin-two propagators use the mass, width, gauge, and component ordering declared by the normalized model. A custom UFO propagator is accepted only when its numerator and denominator can be lowered exactly; optimizations that assume a standard propagator are then disabled. Massless spin two is validated by the scalar-gravity model. Massive spin-two support remains experimental because it has not yet been tested across a comparably broad set of models.

13.4 Support boundary

In the table, *conditional* means that the stated exact algebraic condition must be established for the particular model expression. It does not imply acceptance of every algebraically equivalent UFO spelling.

area	**class**	**status**	**boundary**
source	UFO directory / JSON	supported	UFO Python must be trusted; JSON is data-only and both use the same normalized model

area	class	status	boundary
spin	scalar, Dirac, vector	supported	standard wavefunctions, propagators, component orderings, and closures
spin	massless spin two	supported	tensor path validated by the scalar-gravity model
spin	massive spin two	experimental	Fierz–Pauli path is available, but broad model validation is incomplete
spin	UFO ghosts	partial	propagating ghosts require an exactly representable custom propagator and are not broadly qualified
spin	Majorana/FNV, spin 3/2	unsupported	no fermion-number-violating-flow or Rarita–Schwinger current prescription
colour	$1, 3, \bar{3}, 8$ of SU(3)	supported	LC, NLC, and full-colour treatments described in this report
colour	$6, \bar{6}$, extra groups	unsupported	no sextet or multiple-group flow and contraction representation
arity 3	$1, \delta, T, f$ colour	supported	exact colour projection and representable Lorentz components
arity 3	d^{abc}, ϵ, K_6, T_6	unsupported	no supported colour-flow and contraction prescription
arity 4	colour singlet	conditional	exact tensor reconstruction by non-propagating auxiliary currents
arity 4	two-f gauge colour	conditional	one shared adjoint index, trivalent topology, and exact Lorentz reconstruction
arity 4	other coloured tensors	unsupported	no supported auxiliary colour channel
arity > 4	colour-singlet contact	conditional	exact no-pole reconstruction with the original coupling inserted once
arity > 4	coloured contact	unsupported	no general irreducible-channel factorization
Lorentz	normalized tensor basis	supported	every tensor function must normalize with explicit component axes
Lorentz	form factors / unknown functions	unsupported	no portable numerical prescription
propagator	standard $0, \frac{1}{2}, 1, 2$	supported	model mass and width, with explicit component ordering and closure
propagator	custom UFO tensor	conditional	exact component lowering; standard-propagator optimizations are disabled
symmetry	vectorlike/Yang–Mills/reflection	conditional	exact normalized identities over the complete interaction sector used by the process

References

[1] F. A. Berends and W. T. Giele, *Recursive Calculations for Processes with n Gluons*, Nucl. Phys. B **306** (1988) 759–808, doi:10.1016/0550-3213(88)90442-7.

[2] F. Maltoni, K. Paul, T. Stelzer, and S. Willenbrock, *Color-flow decomposition of QCD amplitudes*, Phys. Rev. D **67** (2003) 014026, doi:10.1103/PhysRevD.67.014026, arXiv:hep-ph/0209271.

[3] C. Degrande, C. Duhr, B. Fuks, D. Grellscheid, O. Mattelaer, and T. Reiter, *UFO—The Universal FeynRules Output*, Comput. Phys. Commun. **183** (2012) 1201–1214, doi:10.1016/j.cpc.2012.01.022, arXiv:1108.2040.

[4] L. Darmé et al., *UFO 2.0—The Universal Feynman Output format*, Eur. Phys. J. C **83** (2023) 631, doi:10.1140/epjc/s10052-023-11780-9, arXiv:2304.09883.

[5] R. Frederix and T. Vitos, *Event generation with exponential scaling in multiplicity using AmpliCol*, JHEP **04** (2026) 192, doi:10.1007/JHEP04(2026)192, arXiv:2601.19483.